

Uncertainty-aware risk-value vehicle–UAV scheduling for transmission-line inspection with probabilistic energy prediction

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ARTICLE INFO

Keywords:

Transmission-line inspection
 Vehicle–UAV scheduling
 Energy uncertainty
 Risk-value routing
 Adaptive large-neighborhood search
 GIS-grounded experiments

ABSTRACT

Large-scale transmission-line inspection increasingly relies on coordinated ground vehicles and unmanned aerial vehicles, but routing decisions remain sensitive to uncertain UAV endurance, heterogeneous tower priorities and sparse feasible stopping locations. This paper formulates transmission-line inspection as an uncertainty-aware risk-value vehicle–UAV scheduling problem. The proposed framework combines a physics-informed probabilistic energy predictor, chance-constrained sortie screening, risk-value task prioritization and an adaptive large-neighborhood search procedure. The energy branch is evaluated with both simulation-generated inspection instances and public quadcopter telemetry, while the routing branch is tested on synthetic corridor instances and public GIS-grounded tower geometries.

Computational experiments cover small reference instances, medium and large instances from 30 to 500 towers, energy-prediction accuracy, ablation tests, candidate-stop screening, sensitivity analysis, GIS-grounded cases and stress scenarios. On 100-tower synthetic instances, the full method reduces risk-weighted completion time by 34.39% relative to fixed-parameter ALNS and by 0.76% relative to point-estimate ALNS-PINN; the RWCT paired comparison is Holm-significant, while the 2.40-percentage-point top-risk coverage gain is descriptive after correction. On 500-tower instances, the corresponding risk-time gains are 36.05% and 1.00%, and the top-risk coverage gain remains Holm-significant. In three public GIS-grounded cases using OpenStreetMap tower nodes and NASA POWER weather, the full method obtains the lowest risk-weighted completion time, although the margins and coverage tradeoffs are case-dependent. A telemetry calibration experiment on 209 AirLab quadcopter flights reduces test energy MAE from 3.90 Wh for a constant baseline to 0.89 Wh for a telemetry-weather linear model. The evidence supports an algorithmic and GIS-grounded scheduling contribution, while field validation with utility inspection records remains future work.

1 Introduction

Transmission-line inspection is becoming a larger operational bottleneck for power-grid resilience. Long overhead corridors are exposed to wind, vegetation, lightning, icing, heat and aging components, while inspection resources must be allocated before incipient defects become outages or safety events. Reviews of power-line inspection show that unmanned aerial systems can improve access to towers, conductors and right-of-way components, but they also introduce perception, navigation, endurance and data-quality constraints that are different from conventional ground patrols (Nguyen et al., 2018; Li et al., 2021b; Ahmed et al., 2024). Recent surveys further indicate that automated inspection is moving from isolated visual detection toward integrated planning pipelines in which sensing, route execution and defect prioritization must be coordinated (Liu et al., 2020; Faisal et al., 2025). These features make transmission-line inspection not only a perception task, but also a transportation scheduling problem in which high-risk assets should be visited early under limited mobility and energy resources.

A practical inspection operation often combines ground vehicles and unmanned aerial vehicles (UAVs). Ground vehicles move crews, batteries and support equipment along accessible roads, while UAVs are launched from feasible stopping points to inspect nearby towers or line segments. This operating mode resembles truck-drone routing, but the objective is not parcel delivery: tower tasks have heterogeneous risk and value, launch and recovery locations are constrained by corridor geometry, and UAV energy depends on weather, payload, service time and the distance between a tower and its candidate stop. Classical truck-drone models established the basic synchronization problem for vehicle-assisted UAV operations (Murray and Chu, 2015; Agatz et al., 2018; Wang and Sheu, 2019; Murray and Raj, 2020), and later work developed exact, branch-and-bound, dynamic-programming and matheuristic methods for several vehicle-drone routing variants (Poikonen et al., 2019; Roberti and Ruthmair, 2021; Bouman et al., 2018; Ha et al., 2018; Schermer et al., 2019). However, inspection scheduling requires a stronger connection between task priority, energy feasibility and corridor-specific stop generation than is needed in many delivery formulations.

Recent routing studies show that the transportation-research literature is already moving toward richer vehicle-drone systems. Examples include energy supply logistics with uncrewed aerial and ground vehicles (Kim et al., 2026), eVTOL-drone routing in continuous spaces with no-fly zones (Liu et al., 2026), air-ground multimodal planning (Zhang et al., 2026), automatic heuristic design for vehicle-drone routing (Shi and Zhen, 2026), and synchronized truck-drone order splitting (Yang and Li, 2025). In parallel, electric-vehicle and drone-routing research has studied time windows, recharging, public transport capacity and other energy-related operational constraints (Schneider et al., 2014; Desaulniers et al., 2016; Moadab et al., 2022). These papers demonstrate the importance of integrated energy and routing decisions, yet most are not designed around inspection-specific risk values or probabilistic UAV endurance.

Energy uncertainty is a central reason why a deterministic endurance model can be misleading in inspection planning. Empirical and modeling studies show that UAV energy consumption varies with payload, speed, wind, temperature, routing geometry and vehicle type (Stolaroff et al., 2018; Zhang et al., 2021; Aiello et al., 2021; Dukkanci et al., 2021; Jeong et al., 2019), and broader drone-delivery assessments show that operational distance and energy assumptions strongly affect environmental and cost conclusions (Goodchild and Toy, 2018; Figliozzi, 2017). Physics-informed and uncertainty-aware learning methods provide a way to combine mechanistic structure with data-driven residual correction (Raissi et al., 2019; Karniadakis et al., 2021; Yang et al., 2021), but their scheduling value depends on how the predicted uncertainty is used by the optimizer. A route that is nominally feasible under mean energy may still violate a battery reserve under wind or payload perturbations. Conversely, an overly conservative route can delay high-risk towers and reduce the operational value of early inspection. This creates a coupled problem: the scheduler must jointly decide which high-value towers to prioritize, which candidate stops to use, and which sorties remain feasible under probabilistic energy bounds.

This paper formulates transmission-line inspection as an uncertainty-aware risk-value vehicle-UAV scheduling problem. We develop a probabilistic energy prediction layer that returns mean and upper-quantile energy estimates, a chance-constrained sortie-screening rule, and a risk-weighted completion objective that prioritizes towers with higher defect risk and asset value. The solution method is an adaptive large-neighborhood search (ALNS) framework built on the destroy-repair and adaptive-weight principles of routing metaheuristics (Ropke and Pisinger, 2006; Sacramento et al., 2019; Zhang et al., 2023), with simulated-annealing acceptance used to diversify the search (Kirkpatrick et al., 1983). Unlike a generic truck-drone delivery model, the proposed framework explicitly uses corridor candidate stops, probabilistic energy feasibility and risk-value task ordering for infrastructure inspection.

The main contributions are as follows.

- We formulate a vehicle-UAV transmission-line inspection scheduling problem that couples tower coverage, candidate stopping points, UAV energy feasibility and risk-weighted inspection priority.
- We introduce a probabilistic energy-prediction and chance-screening layer that passes upper-quantile UAV energy estimates to the routing solver instead of relying on fixed endurance.
- We design an ALNS-based scheduling framework that combines uncertainty-aware feasibility filtering with risk-value task prioritization for medium and large corridor instances.
- We evaluate the framework using proposal-scale synthetic instances, public GIS-grounded tower geometries from OpenStreetMap and HIFLD data (OpenStreetMap contributors, 2026; Homeland Infrastructure Foundation-Level Data, 2026), public weather from NASA POWER (NASA Langley Research Center, 2026), and a public quadcopter telemetry dataset for energy calibration (Rodrigues et al., 2021).

The evidence should be interpreted with a clear boundary. The experiments support an algorithmic and GIS-grounded scheduling contribution, and the energy branch is calibrated against public quadcopter telemetry. They do not constitute field validation on utility-provided inspection missions with real defect labels, crew records or battery logs. The remainder of the paper reviews related vehicle-UAV and inspection-routing studies, defines the optimization model, presents the probabilistic ALNS framework, and reports the computational evidence across synthetic, GIS-grounded and telemetry-calibrated experiment blocks.

2 Related work

This section follows the modeling pipeline used in Sections 3 and 4. The review is organized around three questions: how ground vehicles and UAVs are synchronized, how uncertain energy feasibility enters a routing model, and how an adaptive search procedure

can exploit the resulting service graph for infrastructure inspection. This structure keeps the literature review connected to the formulation rather than treating vehicle-UAV routing, energy modeling, inspection sensing and metaheuristics as independent topics.

2.1 Vehicle-UAV routing for infrastructure inspection

Vehicle-UAV routing supplies the structural starting point for this paper: the inspection plan must decide where a ground vehicle stops, which UAV sorties are launched, and how launch-service-recovery timing is synchronized. The flying sidekick traveling salesman problem introduced this basic coupling for a vehicle that launches a UAV, lets it serve a task and later recovers it (Murray and Chu, 2015). Later work extended the idea to vehicle-routing settings, multiple drones and exact or dynamic-programming solution methods (Agatz et al., 2018; Wang and Sheu, 2019; Roberti and Ruthmair, 2021). These studies justify the route variables, assignment variables and timing constraints used later in the model, because any vehicle-supported inspection scheduler needs the same synchronization backbone.

This backbone is necessary but not sufficient for transmission corridors. Recent drone-assisted routing studies include multiple flying sidekicks, matheuristics, public-transport charging capacity, synchronized order splitting and air-ground multimodal transport networks (Murray and Raj, 2020; Schermer et al., 2019; Yang and Li, 2025; Zhang et al., 2026). These extensions are relevant because they move beyond a single truck and a single drone, but their service nodes are still usually customers, parcels or passenger-demand points. A support vehicle in transmission-line inspection is instead positioning a crew near sparse road-access points so that UAVs can inspect linearly arranged infrastructure assets.

Transmission-line inspection research explains why those assets cannot be treated as anonymous delivery customers. UAV navigation, lidar-assisted sensing and vision-based defect analysis have made aerial inspection practical for long linear infrastructure (Hui et al., 2019; Guan et al., 2021; Li et al., 2021b). Reviews of visual and deep-learning-based power-line inspection show a broader shift from manual patrols toward automated sensing and component-level defect recognition (Nguyen et al., 2018; Liu et al., 2020; Faisal et al., 2025). However, this sensing literature usually asks whether a UAV can acquire useful imagery, navigate safely or identify defects; it less often asks how a ground crew should schedule road-access stops and UAV sorties when many towers compete for limited vehicle time and battery capacity. The first gap is therefore a transport model that keeps vehicle-UAV synchronization, but reinterprets the service nodes as risk-bearing inspection tasks.

2.2 Energy uncertainty and risk-value scheduling

Energy-aware routing is the second foundation because synchronization decisions are meaningful only when the corresponding aerial service edges can actually be flown. Electric-vehicle routing provides the broader precedent for representing range, recharging and time-window restrictions as routing constraints (Schneider et al., 2014; Desaulniers et al., 2016). UAV energy studies then show why a fixed distance threshold is too crude for aerial service: consumption can depend on payload, speed, distance, hovering, route geometry and environmental conditions (Dorling et al., 2017; Dukkanci et al., 2021; Zhang et al., 2021). These findings map directly to the stop-tower pair (s, i) in this paper, because a tower may be geographically close to a candidate stop but still fragile under wind, dwell-time or payload conditions.

Energy modeling also affects the interpretation of routing results. Drone-delivery sustainability studies show that energy and emissions conclusions are sensitive to assumptions about routing, payload and operating context (Stolaroff et al., 2018; Goodchild and Toy, 2018; Figliozzi, 2017). Payload-dependent truck-drone routing further shows that sortie feasibility and route design can interact rather than remain separable preprocessing steps (Jeong et al., 2019). For transmission-line inspection, this interaction appears as a service-graph construction problem. If energy is underestimated, the optimizer may select unsafe sorties; if it is made too conservative without regard to task priority, the schedule may delay high-value towers.

The scheduling layer also needs a reason for serving some feasible towers earlier than others. Inspection value is heterogeneous: a tower with higher defect likelihood, criticality or asset value should not be treated as interchangeable with a low-priority tower simply because both require one service visit. Robust and chance-constrained optimization provide a natural decision logic for conservative service-pair screening (Bertsimas and Sim, 2004; Nemirovski and Shapiro, 2007), while physics-informed and probabilistic learning motivate using uncertainty-aware energy estimates rather than mean predictions alone (Raissi et al., 2019; Karniadakis et al., 2021; Yang et al., 2021). The second gap is therefore the interface between energy uncertainty and priority-aware scheduling: the upper-quantile energy rule should decide which stop-tower edges are admissible, and tower risk-value should decide how the admissible tasks are ordered.

2.3 Adaptive search for integrated service graphs

The preceding two streams define the structure of the problem, but they do not by themselves provide a scalable solution method. Vehicle-UAV synchronization, road-access stop selection, q95 service-pair screening and risk-weighted completion order create a combinatorial scheduling problem whose exact route-sortie formulation grows quickly with the number of towers, stops and UAVs. Exact models remain useful for small reference instances, but medium and large corridor cases require a heuristic that can manipulate task order, stop assignment and feasibility repair without rebuilding a full MILP at every move.

ALNS is well suited to this role because its destroy and repair operators can be specialized to the vehicle route, the feasible service graph and the inspection objective, while its adaptive weights learn which operators are useful during the run. The original

Table 1: Positioning of this study relative to representative vehicle-UAV, energy-aware routing and inspection-planning literature.

Reference	V-UAV	Energy	UQ	Risk/value	Stops	Public data	Main role in the literature
Murray and Chu (2015)	✓	—	—	—	—	—	Launch-recovery synchronization for truck-drone delivery.
Chung et al. (2020)	✓	partial	—	—	partial	—	Review of drone and drone-truck optimization models and research directions.
Li et al. (2021a)	✓	partial	—	—	partial	—	Two-echelon perspective on ground-vehicle and UAV routing families.
Sacramento et al. (2019)	✓	partial	—	—	—	—	ALNS mechanics for drone-assisted vehicle routing.
Dukkanci et al. (2021)	UAV	✓	—	—	—	—	Speed-dependent energy and cost optimization for range-limited drone delivery.
Yang et al. (2023)	✓	partial	robust	—	—	—	Robust drone-truck route planning under travel-time uncertainty.
Moadab et al. (2022)	✓	✓	—	—	partial	—	Vehicle-drone routing with transport-capacity and energy-related constraints.
Kim et al. (2026)	✓	✓	—	—	✓	✓	Two-echelon energy-supply location-routing with resource allocation and wildfire case.
Shi and Zhen (2026)	✓	partial	—	—	—	—	LLM-assisted heuristic design for vehicle-drone collaborative routing.
Inspection sensing (Nguyen et al., 2018; Li et al., 2021b; Liu et al., 2020)	UAV	partial	—	partial	—	real sensing	Transmission-line defect detection, navigation and sensing rather than dispatch optimization.
This work	✓	✓	✓	✓	✓	GIS+telemetry	Probabilistic energy screening and risk-value ALNS for transmission-line inspection scheduling.

ALNS framework introduced adaptive operator selection for large routing problems (Ropke and Pisinger, 2006). Drone-routing applications show that ALNS can handle vehicle-drone synchronization while remaining computationally practical for richer variants (Sacramento et al., 2019; Zhang et al., 2023). For the present paper, the important point is not ALNS as a generic metaheuristic, but the way operator scores can be made sensitive to q95 energy margins, segment workload and risk-value priority.

The missing connection across the literature is a complete loop in which a conservative energy predictor constructs feasible stop-tower edges, inspection risk-value scores guide insertion order, and the schedule is evaluated with inspection-specific metrics rather than makespan alone. Table 1 summarizes this position. Existing vehicle-UAV routing supplies launch-recovery synchronization, energy-aware routing supplies consumption and endurance logic, inspection research supplies the infrastructure-service motivation, and adaptive search supplies scalability. None of these streams alone provides a reproducible scheduling framework that couples road-access candidate stops, upper-quantile UAV energy, tower risk-value priority and ALNS. This is why the model formulation in the next section first defines the corridor service graph, then defines q95 feasibility, then embeds risk-value priority in the objective, and finally prepares the state representation used by the algorithm.

3 Problem description and model formulation

The inspection problem is not a direct copy of parcel-delivery truck-drone routing. A transmission corridor has sparse road access, linearly arranged assets, weather-sensitive UAV sorties and tower-specific risk-value priority. The model therefore has to define not only a vehicle route and UAV assignments, but also the energy-qualified service graph on which those assignments are allowed. Figure 1 summarizes this modeling interface: a crew vehicle visits road-access stops, UAVs inspect towers, wind affects q95 energy, and the retained stop-tower edges enter a risk-value schedule.

The formulation is organized in the same order as the implemented pipeline. Section 3.1 defines the corridor network, inspection tasks and decision variables. Section 3.2 converts the energy predictor into a q95-feasible service graph. Section 3.3 defines the risk-value scheduling objective and the metrics used to evaluate it. Section 3.4 gives the compact MILP reference used for S-scale validation. Section 3.5 then states the boundary between the compact exact reference and the order-based ALNS implementation used for larger cases.

3.1 Operational network and inspection tasks

The inspection system is represented by a corridor network, a set of tower tasks and a set of road-access support stops. Let $\mathcal{T}$ be the set of towers requiring inspection, $\mathcal{S}$ the set of feasible ground-vehicle stops and $\mathcal{S}_0 = \mathcal{S} \cup \{0\}$ the stop set including the depot. A tower task is described by

$$\xi_i = (p_i, s_i, q_i, r_i, v_i, g_i), \quad i \in \mathcal{T}, \quad (1)$$

where $p_i = (x_i, y_i)$ is the tower coordinate, s_i is inspection service time, q_i is a workload or payload proxy, r_i is a defect-risk proxy, v_i is inspection value and g_i is the corridor segment index. The segment index is not a decision variable; it is used by the simulator and ALNS operators to avoid pathological concentration of work in one corridor section.

Ground-vehicle movement is defined on a directed support-stop graph

$$G^g = (\mathcal{S}_0, \mathcal{A}^g), \quad \mathcal{A}^g \subseteq \mathcal{S}_0 \times \mathcal{S}_0, \quad (2)$$

with road distance d_{ab}^g and road travel time $t_{ab}^g = d_{ab}^g / u^g$ for arc $(a, b) \in \mathcal{A}^g$. UAV service is defined on a bipartite stop-tower layer. For stop $s \in \mathcal{S}$ and tower $i \in \mathcal{T}$, the round-trip flight distance is

$$d_{si} = 2\|p_s - p_i\|_2. \quad (3)$$

Table 2: Main notation for the uncertainty-aware vehicle-UAV inspection model.

Category	Symbol	Meaning
Sets and graphs	$\mathcal{T}, \mathcal{S}, \mathcal{S}_0, \mathcal{A}^s$	Towers, candidate stops, candidate stops plus depot, and directed road arcs.
Fleet resources	$\mathcal{V}, \mathcal{D}_v, B_d, \rho$	Support vehicles, UAVs carried by vehicle v , UAV battery capacity and reserve ratio.
Task attributes	$P_i, s_i, q_i, r_i, v_i, g_i$	Tower location, service time, workload/payload proxy, defect-risk proxy, inspection value and corridor segment.
Weather and energy	$\omega, \mu_{si}, \sigma_{si}, Q_{si}, \tau_{si}$	Weather state, mean energy, energy uncertainty, upper-quantile energy and sortie duration for stop-tower pair (s, i) .
Service graph	$a_{si}, \mathcal{F}, m_{si}$	Feasibility indicator, q95-feasible stop-tower edge set and normalized energy margin.
Reference MILP variables	$x_{ab}, z_s, u_s, y_{si}, C_{\max}$	Compact S-scale route arc, visited-stop indicator, MTZ order variable, tower assignment and makespan proxy.
Schedule variables	$\pi, \alpha_i, \delta_i, S_i, C_i$	ALNS tower order, selected stop, selected UAV, start time and completion time.
Priority weights	ψ_i, χ_i, λ	Risk-weighted completion weight, construction priority score and objective coefficients.

The same-stop launch-recovery assumption is used in the experiments: a UAV launched from stop s inspects tower i and returns to the same stop. This assumption matches the current simulator and public GIS cases; richer launch-at- s , recover-at- s' variants would require a larger time-expanded model and are not claimed here.

The resource state includes a set of support vehicles $\mathcal{V}$, UAVs $\mathcal{D}_v$ carried by vehicle v , UAV battery capacity B_d , reserve ratio ρ , vehicle speed u^g and UAV speed u^a . Weather is represented by

$$\omega = (u^w, \theta^w, T, \eta), \quad (4)$$

where u^w is wind speed, θ^w is wind direction, T is temperature and η is the uncertainty level used by the energy surrogate. The model uses four assumptions consistent with the code and experiments: candidate stops are known before planning; each tower is inspected once; risk and value are exogenous planning inputs rather than learned defect labels; and sortie feasibility is decided by an upper-quantile energy requirement rather than by a fixed endurance constant.

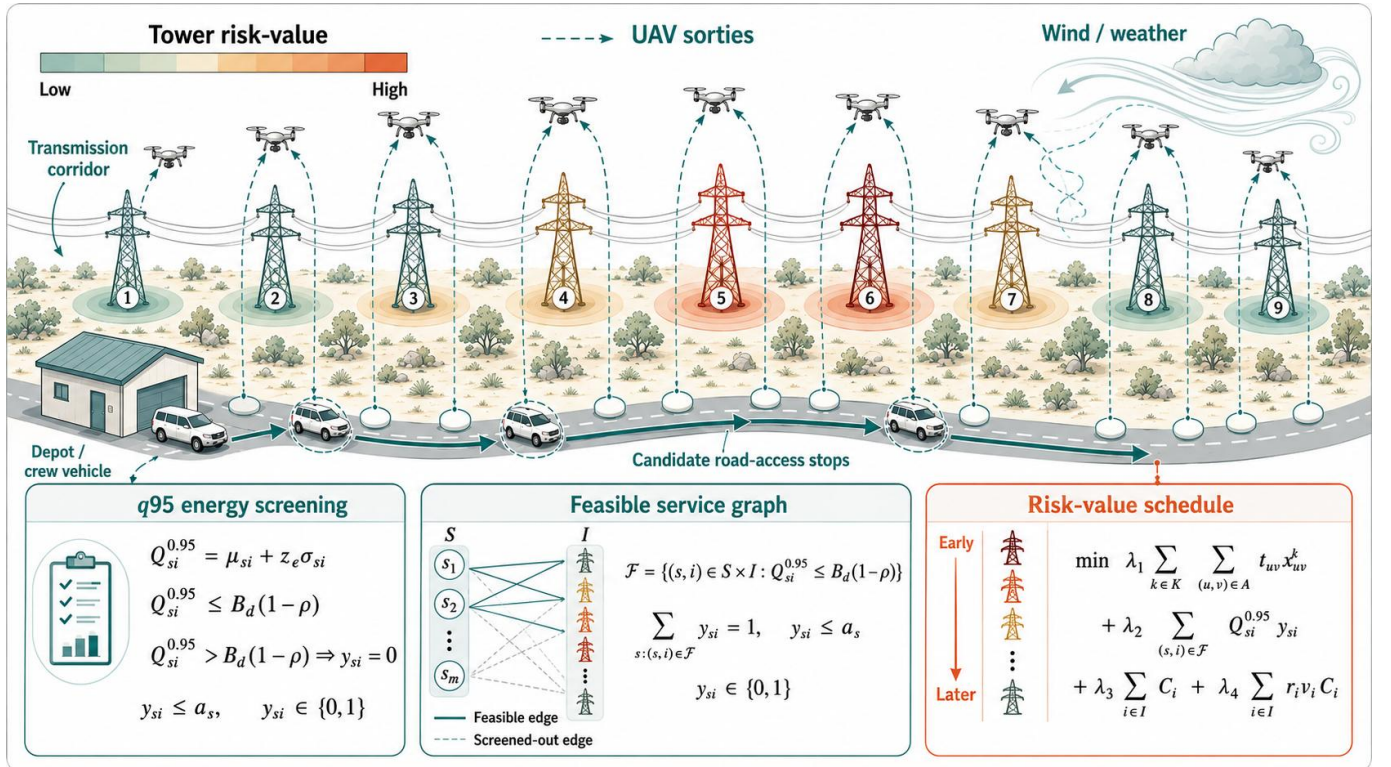


Figure 1: Transmission-line inspection scheduling setting and modeling interface. A crew vehicle travels between road-access candidate stops; UAV sorties inspect towers along a transmission corridor; tower color indicates risk-value priority; wind and weather affect the q95 energy screen. The lower panels show how upper-quantile energy feasibility constructs the stop-tower service graph and how the resulting feasible assignments enter the risk-value schedule.

3.2 q95 service-pair graph

For each stop-tower pair, the energy model decomposes the predicted requirement into flight, hover, sensing, payload, wind and temperature components:

$$E_{si} = E^0 + E_{si}^{fly} + E_i^{hover} + E_i^{sense} + E_{si}^{payload} + E_{si}^{wind} + E_{si}^{temp}. \quad (5)$$

In the implementation this decomposition is evaluated from the round-trip distance, tower service time, payload proxy and weather state. The probabilistic surrogate returns a mean, standard deviation and sortie duration,

$$(\mu_{si}, \sigma_{si}, \tau_{si}) = f_{\theta}(d_{si}, q_i, s_i, B_d, u^a, \omega), \quad (6)$$

where d_{si} is given by Eq. (3). The upper-quantile energy used for screening is

$$Q_{si} = \mu_{si} + z_{\epsilon} \sigma_{si}. \quad (7)$$

The chance-style feasibility rule retains a service edge only when the upper-quantile requirement fits within the battery reserve:

$$Q_{si} \leq B_d(1 - \rho). \quad (8)$$

This rule is not a full stochastic programming model. It is a conservative interface between energy prediction and routing, designed to prevent the optimizer from selecting sorties that are only feasible under mean energy.

The binary service-pair indicator and feasible edge set are

$$a_{si} = \begin{cases} 1, & Q_{si} \leq B_d(1 - \rho), \\ 0, & \text{otherwise,} \end{cases} \quad s \in \mathcal{S}, i \in \mathcal{T}, \quad (9)$$

and

$$\mathcal{F} = \{(s, i) \in \mathcal{S} \times \mathcal{T} : a_{si} = 1\}. \quad (10)$$

The normalized energy margin used by the ALNS operators is

$$m_{si} = \frac{B_d(1 - \rho) - Q_{si}}{B_d}, \quad (s, i) \in \mathcal{S} \times \mathcal{T}. \quad (11)$$

Positive m_{si} indicates a q95-feasible pair; small positive values identify fragile but admissible sorties.

The candidate-stop generator may further restrict the service layer before optimization. Let $\mathcal{K}_i \subseteq \mathcal{S}$ be the screened candidate stops retained for tower i , obtained either by direct enumeration or by corridor clustering. The service set used by the solver is

$$\mathcal{F}^K = \{(s, i) : s \in \mathcal{K}_i, a_{si} = 1\}. \quad (12)$$

If no q95-feasible stop remains for a tower in a stress case, the implementation keeps the least-energy nearest candidate with a large infeasibility penalty for diagnostic purposes. In the q95 ALNS schedule simulator, a near-boundary conservative sortie mode is allowed only when $B_d(1 - \rho) < Q_{si} \leq \chi B_d(1 - \rho)$:

$$\bar{Q}_{si} = \min\{\gamma_Q Q_{si}, (1 - \epsilon_B) B_d(1 - \rho)\}, \quad \bar{\tau}_{si} = \gamma_{\tau} \tau_{si}. \quad (13)$$

For ordinary q95-feasible pairs that do not use this near-boundary mode, $\bar{Q}_{si} = Q_{si}$ and $\bar{\tau}_{si} = \tau_{si}$. Thus barred symbols denote the implemented energy and time quantities passed from the energy layer to the routing layer. The experiments use this rule as a reliability-mitigation proxy, not as field validation of a specific flight-control action. Pairs outside the near-boundary band remain infeasible.

3.3 Risk-value scheduling objective

The inspection-specific objective is based on the idea that delaying a high-risk, high-value tower is worse than delaying a low-risk tower. The evaluation weight used for risk-weighted completion time is

$$\psi_i = r_i v_i \quad (14)$$

The construction priority used by the heuristic is

$$\chi_i = v_i(1 + \beta_r r_i), \quad (15)$$

where β_r controls how aggressively high-risk towers are moved forward during construction and repair. The primary inspection metric reported later is risk-weighted completion time,

$$\text{RWCT} = \sum_{i \in \mathcal{T}} \psi_i C_i. \quad (16)$$

For the operational scheduling target used by the order-based ALNS evaluator, the composite objective is written as

$$\min F(X) = \lambda_1 C_{\max} + \lambda_2 \sum_{(a,b) \in \mathcal{A}^g} t_{ab}^g x_{ab} + \lambda_3 \sum_{i \in \mathcal{T}} \sum_{s \in \mathcal{S}} \bar{Q}_{si} y_{si} + \lambda_4 \sum_{i \in \mathcal{T}} \psi_i C_i + M \sum_{i \in \mathcal{T}} \sum_{s \in \mathcal{S}} (1 - a_{si}) y_{si}. \quad (17)$$

The first two terms control makespan and road travel, the third term penalizes conservative UAV energy, the fourth term is the inspection-priority term and the last term prevents infeasible assignments except in explicitly diagnosed stress cases. The ALNS implementation uses a normalized construction score consistent with Eq. (17), while the results report makespan, energy, infeasible-sortie rate, top-risk coverage and RWCT rather than claiming that a single scalar objective captures all operational preferences. The compact HiGHS reference below optimizes a narrower makespan–energy formulation and does not contain the $\psi_i C_i$ objective term; RWCT and TopCov for the HiGHS row are post-hoc evaluator diagnostics.

3.4 Compact MILP reference model

The full synchronized vehicle-UAV model with multiple support vehicles, detailed launch-recovery timing and time-expanded UAV states is too large for the medium and large experiments. Following the role played by exact models in computational routing papers, this study uses a compact MILP only as an S-scale makespan-energy reference. The compact reference assigns each tower to a selected stop, connects selected stops by one aggregate support tour and lower-bounds the makespan by road travel plus parallel UAV workload. It contains no per-tower completion-time variables C_i and therefore no exact RWCT objective. This is the model implemented in the HiGHS reference solver.

Each tower is assigned once:

$$\sum_{s \in \mathcal{S}} y_{si} = 1, \quad \forall i \in \mathcal{T}. \quad (18)$$

Assignments are allowed only when the stop is activated and the q95 service edge is admissible:

$$y_{si} \leq z_s, \quad y_{si} \leq a_{si}, \quad \forall s \in \mathcal{S}, i \in \mathcal{T}. \quad (19)$$

The selected stops must form a depot-connected support route:

$$\sum_{b \in \mathcal{S}} x_{0b} = 1, \quad \sum_{a \in \mathcal{S}} x_{a0} = 1, \quad (20)$$

and

$$\sum_{b \in \mathcal{S}_0: b \neq s} x_{sb} = z_s, \quad \sum_{a \in \mathcal{S}_0: a \neq s} x_{as} = z_s, \quad \forall s \in \mathcal{S}. \quad (21)$$

MTZ order variables remove disconnected subtours among selected stops:

$$\ell_a - \ell_b + |\mathcal{S}|x_{ab} \leq |\mathcal{S}| - 1, \quad \forall a, b \in \mathcal{S}, a \neq b. \quad (22)$$

Stop-level UAV workload is bounded by count, payload and q95 energy:

$$\sum_{i \in \mathcal{T}} y_{si} \leq \bar{n}z_s, \quad \sum_{i \in \mathcal{T}} q_i y_{si} \leq \bar{q}z_s, \quad \forall s \in \mathcal{S}, \quad (23)$$

and

$$\sum_{i \in \mathcal{T}} \bar{Q}_{si} y_{si} \leq |\mathcal{D}|B_d(1 - \rho)z_s, \quad \forall s \in \mathcal{S}. \quad (24)$$

The workload bound is deliberately compact: it approximates multiple parallel UAV sorties from a selected stop rather than modeling every launch and recovery time exactly. The makespan proxy used in the reference model is

$$C_{\max} \geq \sum_{(a,b) \in \mathcal{A}^g} t_{ab}^g x_{ab} + \frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{T}} \sum_{s \in \mathcal{S}} \bar{\tau}_{si} y_{si}. \quad (25)$$

For S-scale instances this compact model provides a useful numerical reference. For M- and L-scale cases it is not treated as the operational solver.

3.5 Model-to-ALNS interface

The ALNS implementation solves the large cases through an order-based schedule state rather than through the full compact MILP variable set. A candidate solution is represented by a tower permutation π , a selected stop $\alpha_i \in \mathcal{S}$, a UAV resource δ_i and start/completion times (S_i, C_i) . Given a tower order, the scheduler selects a stop by the q95-aware score

$$\alpha_i = \arg \min_{s \in \mathcal{S}} \{ \bar{Q}_{si} + 0.2\bar{\tau}_{si} + P(1 - a_{si}) \}, \quad (26)$$

where P is a large infeasibility penalty. The tower is then assigned to the earliest available UAV:

$$\delta_i = \arg \min_{d \in \mathcal{D}} A_d, \quad S_i = A_{\delta_i}, \quad C_i = S_i + \bar{\tau}_{\alpha_i}, \quad A_{\delta_i} \leftarrow C_i. \quad (27)$$

This is the computational state used by the current Python implementation. Tower value and risk affect tower ordering, removal and insertion scores rather than the stop selector in Eq. (26); otherwise a tower-specific constant would not change the selected stop. The state preserves the key model interfaces-q95 feasibility, stop choice, energy cost, completion time and risk-value priority-while avoiding a full time-expanded synchronized MILP at every neighborhood move.

This boundary is important for interpreting the paper. The mathematical formulation defines the inspection scheduling structure and the compact S-scale reference problem. The ALNS solver uses the same energy, service-graph and risk-value objects, but evaluates moves through the order-based simulator in Eqs. (26)–(27). Consequently, the contribution claimed here is a scalable probabilistic risk-value scheduling framework with a compact makespan-energy reference, not a new exact branch-and-price or full multi-vehicle synchronized MILP for transmission-line inspection.

4 Solution method

The solution method is organized around the computational bottleneck created by the model: the scheduler must choose a vehicle-supported UAV service order while avoiding stop-tower pairs that are only feasible under optimistic energy assumptions. We therefore separate the algorithm into a service-graph construction stage and a risk-value ALNS stage (Fig. 2). The first stage converts the probabilistic energy interface into a sparse q95-feasible graph. The second stage searches over service sequences, stop assignments and completion times on that graph.

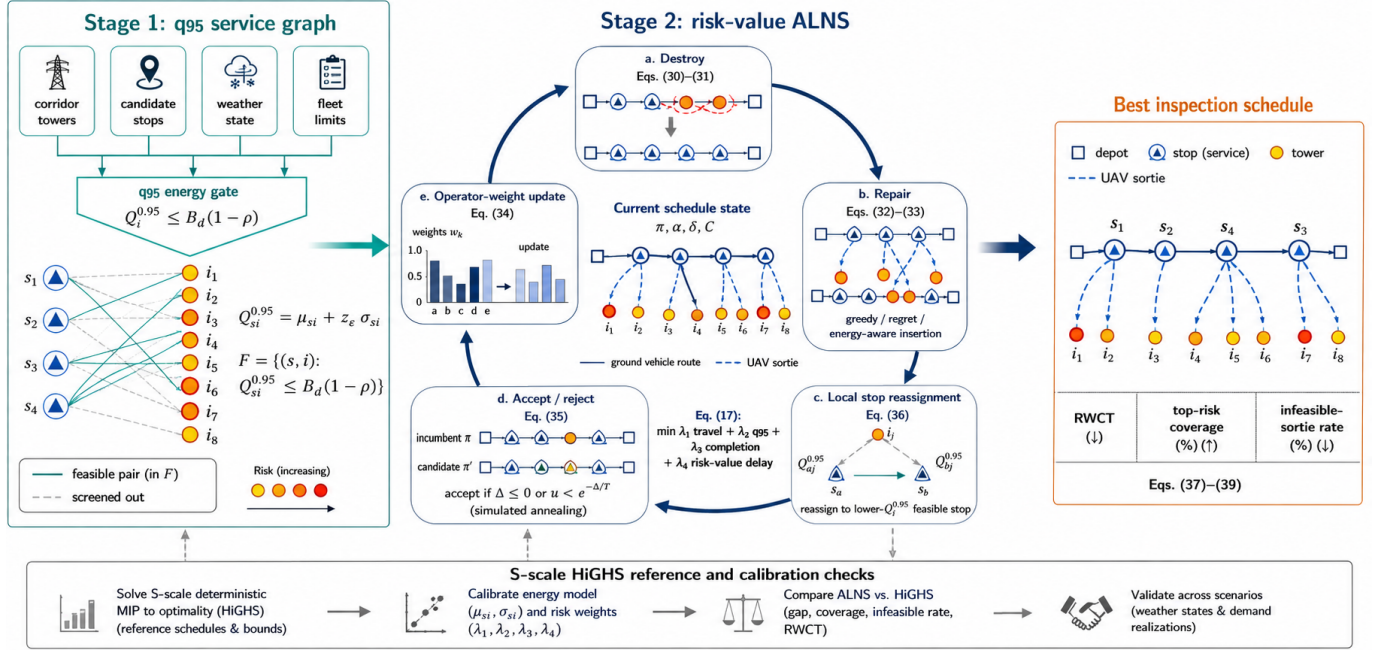


Figure 2: Formula-linked two-stage inspection scheduler. Stage 1 converts corridor towers, candidate stops, weather state and fleet limits into a q95-feasible service graph; Stage 2 improves the current schedule through destroy, repair, local stop reassignment, simulated-annealing acceptance and adaptive operator-weight updating. The output panel summarizes the best risk-value schedule and the evaluation metrics used in Eqs. (37)–(40); the lower band shows the S-scale HiGHS reference and calibration checks.

Figure 2 should be read as the executable link between the formulation and the experiments. The left block implements the chance-feasible service set in Eqs. (7)–(10); the central loop updates the order-based schedule state in Eq. (29); and the right block reports the same RWCT, coverage and infeasibility diagnostics used later in Section 6. This structure prevents the algorithm from becoming a collection of unrelated heuristic operators: every move is evaluated on the q95 service graph and against the risk-value objective. Stage 1 constructs this feasible service layer. It generates candidate stops from corridor geometry, evaluates stop-tower pairs with the probabilistic energy interface, and removes all pairs outside $\mathcal{F}$. Direct enumeration is possible but creates many weak stop-tower pairs. Therefore, K-means and density-based stop generation are tested as screening alternatives, and the default screening mode is selected from the candidate-stop experiment. The output of Stage 1 is a sparse bipartite service graph

$$G_F = (\mathcal{S}, \mathcal{T}, \mathcal{F}; Q_{si}, \tau_{si}, d_{si}, m_{si}), \quad m_{si} = \frac{B_d(1-\rho) - Q_{si}}{B_d}, \quad (28)$$

where m_{si} is the normalized energy margin. Positive margin indicates a chance-feasible service pair, while small positive margin indicates pairs that are feasible but operationally fragile.

Stage 2 searches over inspection schedules. A solution state is represented as

$$x = \{\Pi, a(i), s(i), C_i\}_{i \in \mathcal{T}}, \quad (29)$$

where Π is the tower service order, $a(i)$ is the assigned UAV/vehicle resource, $s(i)$ is the launch stop selected from feasible pairs, and C_i is the resulting completion time. Given an order Π , the stop selector chooses feasible stops by combining route proximity, upper-quantile energy and current stop workload. The evaluator then computes makespan, risk-weighted completion time, energy use, infeasible-sortie rate and top-risk coverage. This representation is computationally light enough for 500-tower instances, but it is less expressive than a full time-expanded exact model; this explains why the repair-operator evidence is interpreted conservatively in Section 6.

The ALNS neighborhood contains destroy and repair operators matched to the inspection setting. Random removal diversifies the search. Worst-energy removal targets towers with high upper-quantile energy or narrow battery margins. For a tower currently assigned to stop $s(i)$, the removal priority is

$$\Omega_i = \omega_1 \frac{Q_{s(i)i}}{B_d(1-\rho)} + \omega_2 (1 - m_{s(i)i}) + \omega_3 \frac{\psi_i}{\max_{j \in \mathcal{T}} \psi_j}. \quad (30)$$

The first two terms target energy-fragile assignments; the third allows high-priority tasks to be reconsidered when the current ordering delays them.

Shaw-related removal removes towers that are similar in corridor position, service feasibility and risk-value. For seed tower i , relatedness to tower j is measured as

$$R_{ij} = \alpha_1 \frac{\|p_i - p_j\|}{\bar{d}} + \alpha_2 \frac{|\psi_i - \psi_j|}{\bar{\psi}} + \alpha_3 \left(1 - \frac{|\mathcal{F}_i \cap \mathcal{F}_j|}{|\mathcal{F}_i \cup \mathcal{F}_j|} \right), \quad (31)$$

where p_i is tower position, $\mathcal{F}_i = \{s : (s, i) \in \mathcal{F}\}$ and lower R_{ij} means stronger relatedness. Path-segment removal changes contiguous corridor sections, while UAV-chain removal removes a group of towers served from the same support stop.

The repair pool includes greedy insertion, regret insertion, energy-minimum insertion and synchronization-aware insertion. For a removed tower i , candidate insertion at position p , stop s and UAV d is scored by

$$I(i, p, s, d) = \Delta F(i, p, s, d) + \beta_1 \frac{Q_{si}}{B_d(1 - \rho)} + \beta_2 \Delta W(s, d) - \beta_3 \frac{\psi_i}{1 + \widehat{C}_i}, \quad (32)$$

where ΔF is the change in the weighted routing objective, ΔW is the induced dwell-time change and $\widehat{C}_i$ is the tentative completion time. Greedy repair selects the feasible insertion with minimum I . Regret repair first computes the ordered insertion scores $I_{i(1)} \leq I_{i(2)} \leq \dots$ and selects the tower with the largest

$$\text{regret}_\kappa(i) = \sum_{r=2}^{\kappa} (I_{i(r)} - I_{i(1)}). \quad (33)$$

Energy-minimum repair emphasizes the second term in Eq. (32); synchronization-aware repair emphasizes ΔW . These operators are implemented as search components rather than claimed as independent theoretical contributions.

Let $\mathcal{O}$ denote the operator pool. Operator weights are updated from accepted and improving moves:

$$\pi_{h+1}^k = (1 - \eta) \pi_h^k + \eta \frac{s_h^k + \epsilon_\pi}{\sum_{\ell \in \mathcal{O}} (s_h^\ell + \epsilon_\pi)}, \quad (34)$$

where π_h^k is the probability of operator k at update epoch h , s_h^k is its accumulated score, ϵ_π prevents zero-score collapse and η is the adaptation rate. The score is increased when an operator produces a feasible solution, an accepted non-improving solution, an improving incumbent or a new global best. A simulated-annealing acceptance rule is used for non-improving moves:

$$P(\text{accept}) = \exp \left[-\frac{F(x') - F(x)}{T_h} \right]. \quad (35)$$

Here $F(x)$ is the weighted objective in Eq. (17) evaluated by the heuristic schedule simulator. The temperature follows $T_{h+1} = \alpha T_h$. After accepted moves, a local search step tests adjacent tower swaps, short segment reversals and stop reassignment for towers whose feasible-stop set has more than one member. A stop reassignment for tower i selects

$$s_i^* = \arg \min_{s \in \mathcal{F}_i} \left\{ \gamma_1 Q_{si} + \gamma_2 d_{s, \text{prev}(i)}^{\text{road}} + \gamma_3 \Delta W(s) \right\}. \quad (36)$$

These local moves are deliberately limited so that the experiment can separate broad uncertainty/risk-value effects from expensive exact synchronization.

Equations (28)–(36) form one algorithmic loop rather than separate heuristic decorations. Eq. (28) defines the feasible search space, Eq. (29) defines the mutable solution state, Eqs. (30) and (31) choose what to destroy, Eqs. (32) and (33) choose how to repair, Eq. (36) performs a local feasible-stop move, and Eqs. (35) and (34) decide whether the move is retained and how future operators are sampled.

The three mechanisms most specific to inspection scheduling are service-pair screening, risk-value destroy–repair and feasible-stop reassignment. They are summarized in Fig. 2 and should be read with Algorithm 1: the q95 gate defines the retained edges before ALNS begins, destroy–repair acts during each iteration, and local reassignment is applied only when a tower has more than one feasible support stop. This avoids treating the operators as isolated tricks; each move is tied to the feasible service graph, objective in Eq. (17) and implemented schedule state in Eqs. (26)–(27).

The stress experiment is deliberately used to restrict the contribution claim. It shows that the full uncertainty/risk-value framework is useful under difficult conditions, but it does not show that energy-minimum and synchronization-aware repair operators independently improve final aggregate quality in the current order-only ALNS implementation. They are therefore described as search components and diagnostics, not as independent headline innovations.

5 Computational experiments

The experiments are designed to test the modeling and algorithmic choices introduced in Sections 3 and 4. Small instances check whether the compact formulation, q95 service graph and heuristic evaluator operate on compatible scales. Medium and large synthetic instances test the risk-value ALNS against controlled baselines. Energy-prediction experiments test the probabilistic interface before it is used for routing. Stress, public-GIS and screening blocks then test whether the same design remains interpretable under adverse feasibility, external geometry and graph-sparsity changes.

Algorithm 1 Probabilistic risk-value ALNS for vehicle-UAV inspection

Require: Towers $\mathcal{T}$, stops $\mathcal{S}$, weather (ω, T) , UAV parameters (B_d, ρ) , weights λ and iteration budget H .

Ensure: Best schedule x^* , objective trace, feasibility diagnostics and figure/table source data.

- 1: Evaluate $(\mu_{si}, \sigma_{si}, \tau_{si})$ for each stop-tower pair using Eq. (6).
- 2: Compute Q_{si} and build $\mathcal{F}$ from Eqs. (7)–(10).
- 3: Construct G_F and margins m_{si} by Eq. (28).
- 4: Initialize $x = \{\Pi, a(i), s(i), C_i\}$ from Eq. (29) with a risk-value ordered feasible assignment.
- 5: **for** $h = 1, \dots, H$ **do**
- 6: Select destroy and repair operators according to π_h^k in Eq. (34).
- 7: Remove towers using random, path, UAV-chain, Ω_i in Eq. (30) or R_{ij} in Eq. (31).
- 8: Reinsert removed towers by minimizing $I(i, p, s, d)$ in Eq. (32); use Eq. (33) when regret repair is selected.
- 9: Apply adjacent swaps, short segment reversals and feasible-stop reassignment by Eq. (36).
- 10: Recompute C_i , $C_{\max}$, objective value in Eq. (17), RWCT and infeasible-sortie diagnostics.
- 11: Accept x' if it improves $F(x)$ or passes Eq. (35); update x^* if the incumbent improves.
- 12: Update operator scores, π_h^k by Eq. (34) and temperature T_h .
- 13: **end for**
- 14: **return** x^* and diagnostics.

5.1 Instances and data sources

The instance design separates reproducible synthetic control from public-source grounding (Table 3). Synthetic corridor instances follow the proposal scale. S-scale instances contain 10, 15 and 20 towers with 5, 8 and 10 candidate stops and are used only for compact MILP/HiGHS reference checks. M-scale instances contain 30, 50, 75 and 100 towers and support algorithm comparison, ablation and candidate-stop screening. L-scale instances contain 150, 200, 300 and 500 towers and support scalability analysis. Synthetic instances use ten random seeds per main parameter combination. Tower coordinates are generated along two to four perturbed transmission corridors; candidate stops are generated from simplified road-access geometry or from clustering modes that emulate the K-means/DBSCAN stop-generation step. The same nominal “50 towers/25 stops/K-means” label appears in several tables, but the blocks differ by purpose and uncertainty setting: P2 and P6 use medium-uncertainty comparison/screening settings, whereas P4 and P8 use high-uncertainty ablation/sensitivity settings.

Table 3: Instance blocks, data attributes and evidence boundaries used in the computational study.

Block	Scale	Observed or generated inputs		Experimental use		Boundary
		Source	Key fields	Controlled fields	Result blocks	
Synthetic corridors	S/M/L	Seeded corridor generator	Tower x, y ; corridor segment; candidate stops; weather state	Risk; value; service time; payload; stop-generation mode	S-scale reference; algorithm comparison; ablation; screening; sensitivity	Controlled simulation, not utility work orders
Public GIS cases	3 regions	OpenStreetMap towers, HIFLD line context and NASA POWER weather (OpenStreetMap contributors, 2026; Homeland Infrastructure Foundation-Level Data, 2026; NASA Langley Research Center, 2026)	Projected tower geometry; local weather; 60 towers; 30 stops	Proxy risk/value labels; generated road-access stops	GIS map; GIS baseline table; operating-state diagnostic	Public-data grounded geometry; no field-operation claim
AirLab telemetry	209 flights	Public AirLab package-delivery UAV telemetry (Rodrigues et al., 2021)	Route variables; weather variables; measured energy	Train/test split; model family; calibration features	Energy table; energy figure; scheduler-interface check	UAV energy calibration only, not line-inspection telemetry
Stress/screening variants	M/L	Perturbed synthetic generator and candidate-stop modes	Wind; battery reserve; stop density; feasible-pair count	Sparse high wind; tight battery; DBSCAN/K-means/direct; UAV count	Stress tests; scalability; response surface	Mechanism robustness under controlled adverse settings

Table 4: Experimental blocks and traceability from design variables to reported evidence.

Result block	Instance design		Compared factors		Reported evidence	
	Scale	Seeds	Fixed settings	Varied or compared	Primary metrics	Evidence target
S-scale reference	10–20 towers	10	Shared service graph; HiGHS-compatible compact model	HiGHS; point ALNS-PINN; proposed	Makespan; post-hoc RWCT/top coverage; IFR	Model-evaluator consistency, not RWCT optimality
Main algorithm comparison	30–500 towers	10	$\rho = 0.12$; $z_\epsilon = 1.645$; K-means stops	Nearest; GA; ACO; fixed ALNS; point/UQ/full ALNS-PINN	RWCT; top coverage; IFR; runtime	Value of UQ/risk-value scheduling over generic routing
Energy prediction	10 seeds; 209 flights	Split	Same energy-feature definitions	Fixed physics; point/probabilistic PINN; telemetry linear models	MAE; RMSE; MAPE; 95% coverage; false feasible	Whether energy uncertainty is reliable enough for q95 screening
Ablation and stress	50 towers; adverse cases	10	Same ALNS budget within each block	No PINN; no UQ; no risk; no adaptive; no clustering; no repair; stress removals	RWCT; top coverage; IFR; accepted/improving moves	Mechanism contribution and adverse feasibility behavior
Public GIS cases	60 towers/30 stops	3 seeds	Public geometry/weather; proxy labels	Same seven routing methods	RWCT; top coverage; IFR; local map	External geometry grounding without field-validation claim
Scalability/screening	30–500 towers	10	Proposed method unless varied	Tower count; stop mode; iterations; z_ϵ ; ρ ; UAV count	Runtime; pair reduction; feasible pairs; RWCT; IFR	Stage-1 graph density and operational parameter sensitivity

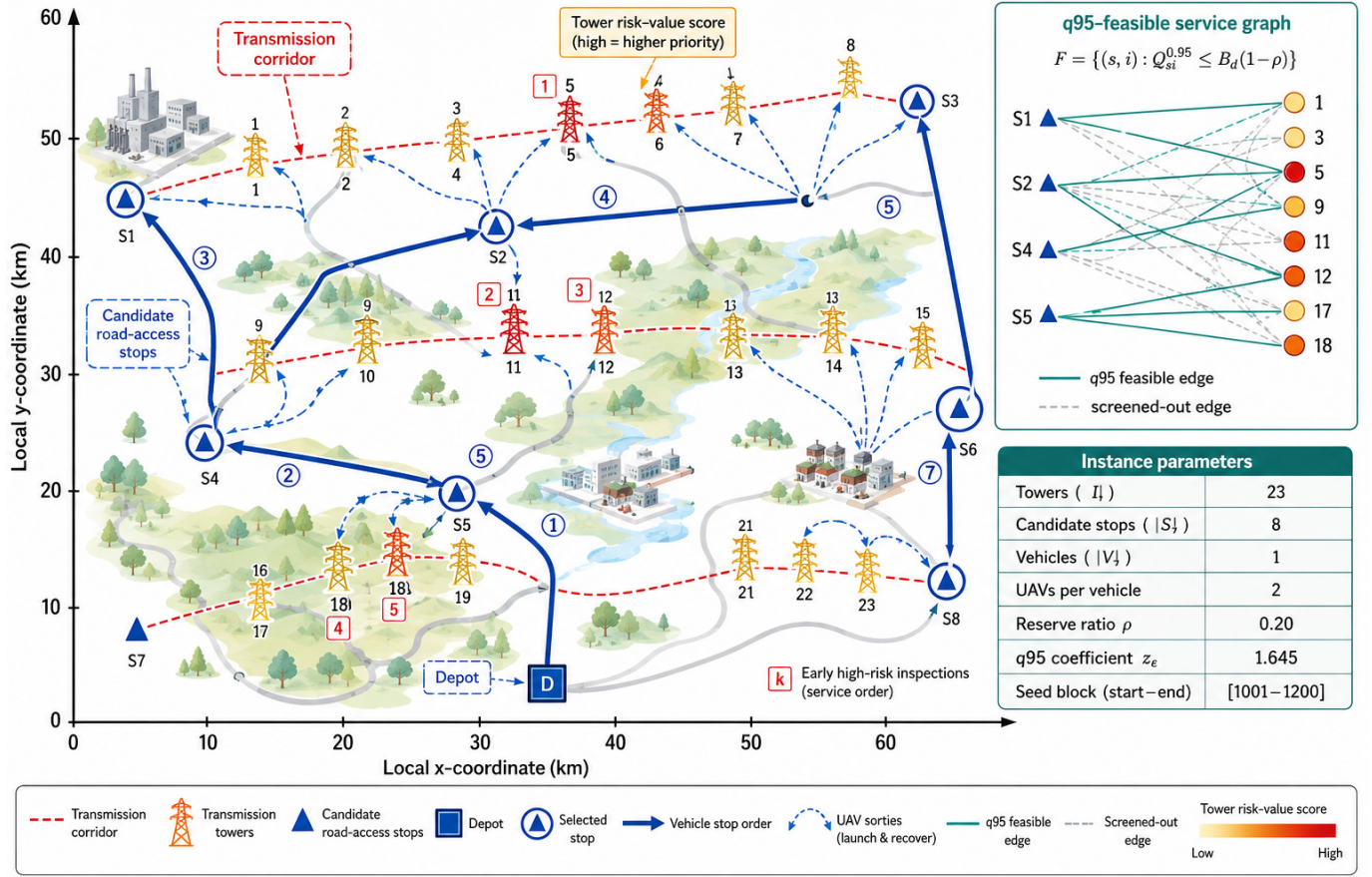


Figure 3: Representative synthetic corridor and dispatch representation used for the controlled experiments. Towers are colored by risk-value priority; candidate road-access stops and the depot define the support network; solid blue arrows show the selected vehicle stop order and dashed blue arcs show UAV launch–recovery sorties. The insets report the q95-feasible service graph and the instance parameters used to connect the generated corridor data to the scheduler.

Figure 3 is included before the numerical results to fix the common experimental language. The tower colors correspond to the priority weights ψ_i used in RWCT; the q95 service-graph inset corresponds to the feasible-pair set $\mathcal{F}$; and the vehicle/UAV routes correspond to the implemented schedule state in Eqs. (26)–(27). Thus the later tables are not abstract benchmark scores alone: they measure how changing energy uncertainty, risk priority or candidate-stop screening changes this dispatch structure.

The public GIS batch uses three regions built from OpenStreetMap tower nodes, HIFLD transmission-line context and NASA POWER weather (OpenStreetMap contributors, 2026; Homeland Infrastructure Foundation-Level Data, 2026; NASA Langley Research Center, 2026). These cases use real public geometry and weather, but tower risk, inspection value and candidate-stop attributes are generated proxies. The energy-calibration block uses 209 public AirLab package-delivery quadcopter flights (Rodrigues et al., 2021). That telemetry is real UAV data, but it is not transmission-line inspection telemetry.

Unless a sensitivity run states otherwise, the energy model uses reserve ratio $\rho = 0.12$ and quantile coefficient $z_\epsilon = 1.645$. The publishable experiment driver uses 80, 100, 60 and 80 ALNS iterations for S, M, L and case-study runs, respectively; GA and ACO use half of the corresponding budget with a minimum of 40 iterations. The repair-stress experiment uses 140 iterations and tighter reserve/quantile settings to create adverse feasibility conditions. These budgets are intended to compare algorithmic behavior under a common local Python implementation, not to benchmark optimized production solvers.

5.2 Baselines

The baseline set is organized as a controlled design ladder rather than a list of unrelated algorithms (Table 5). The first three methods test whether generic order construction or generic metaheuristic routing already explains the results. Fixed ALNS then keeps the same destroy–repair search family but removes the two manuscript-specific layers. The point and UQ ALNS-PINN variants isolate the energy-prediction and conservative-screening effects before the full UQ-RV-ALNS configuration is evaluated.

This structure is important for interpretation in Section 6. A large gap over nearest, GA or ACO would only show that the problem benefits from a stronger routing heuristic. A gap over fixed ALNS tests the value of departing from fixed endurance and cost-only priority. A gap over point or UQ ALNS-PINN is narrower but more relevant: it asks whether upper-quantile feasibility and risk-value scheduling still matter after the solver already uses the same ALNS infrastructure.

Table 5: Baseline algorithms used to separate generic routing competence from the proposed energy-uncertainty and risk-value mechanisms.

Family	Method label	Reference role	Search object	q95 energy	Risk-value	Selection reason
Constructive	Nearest	TSP constructive heuristic (Rosenkrantz et al., 1977)	Spatial tower order; nearest support stop	No	No	Low-complexity spatial baseline
Population	GA	Evolutionary search (Holland, 1975)	Tower-order permutation	No	Indirect	Generic population metaheuristic
Pheromone	ACO	Routing sequence search (Dorigo and Gambardella, 1997)	Tower-order permutation	No	Indirect	Generic routing metaheuristic with path memory
Destroy-repair	Fixed ALNS	ALNS family (Ropke and Pisinger, 2006; Sacramento et al., 2019)	Order and stop-assignment repairs	No	No	Same local-search infrastructure without paper-specific layers
Energy model	Point ALNS-PINN	Internal controlled variant	Mean-energy order and stop choice	No	Partial	Tests whether point prediction alone explains the gain
Uncertainty	UQ ALNS-PINN	Internal controlled variant	q95 screened service graph	Yes	Partial	Isolates conservative feasibility from full risk-value configuration
Proposed	UQ-RV-ALNS	This paper	q95 service graph plus adaptive risk-value ALNS	Yes	Yes	Full configuration tested against all controls

5.3 Evaluation metrics

The evaluation does not rely on a single objective value, because the model itself couples time, priority, energy feasibility and service-graph sparsity. Table 6 lists the metric groups, implementation fields and selection logic. The key point is that each reported metric corresponds to a model or algorithmic decision: RWCT evaluates the risk-value objective, infeasible-sortie rate evaluates the q95 gate, and candidate-pair reduction evaluates the Stage-1 service graph before ALNS search begins.

Table 6: Metric groups, code fields and selection logic used in the computational study.

Metric group	Reported metric	Formula or implementation field	Direction	Used in	Selection logic
Operational efficiency	Makespan; vehicle distance; runtime	Evaluator fields: makespan, distance and runtime	Down	Tables 8, 9, 13	Captures time, ground support movement and computational cost
Inspection priority	RWCT	Eq. (37); evaluator RWCT field	Down	Routing/stress/GIS blocks; post-hoc in S-scale reference	Directly matches the ALNS risk-value term; diagnostic only for compact HiGHS rows
Inspection priority	Top-risk and feasible top-risk coverage	Eqs. (38)–(39); evaluator fields	Up	Algorithm, stress, GIS and screening blocks	Separates nominal early service from early service that is also q95-feasible
Reliability	Infeasible-sortie rate	Eq. (40); evaluator IFR field	Down	Algorithm, stress, GIS, screening and sensitivity blocks	Tests whether q95 feasibility changes decisions, not only reported scores
Energy prediction	MAE; RMSE; MAPE; 95% coverage; false feasible	Prediction-summary CSV fields	Mixed	Energy-evidence block	Separates accuracy, calibration and decision-side false feasibility
Service graph	Feasible pairs; candidate-pair reduction	Feasible-pair count and pair-reduction ratio	Mixed	Scalability and screening blocks	Measures Stage-1 graph sparsity before ALNS sees the instance
Statistical evidence	Paired Wilcoxon; Holm correction	P7 paired seed tests (Wilcoxon, 1945; Holm, 1979)	By metric	Main matched-seed comparisons	Avoids treating repeated seeded runs as unrelated samples

Risk-weighted completion time is

$$\text{RWCT} = \sum_{i \in \mathcal{T}} \psi_i C_i, \quad (37)$$

and measures whether high-risk, high-value towers are inspected early. For the highest-priority subset $\mathcal{T}_\alpha$ and early-service cutoff

θ_{early} , top-risk coverage is

$$\text{TopCov} = \frac{1}{|\mathcal{T}_\alpha|} \sum_{i \in \mathcal{T}_\alpha} \mathbf{1}\{C_i \leq \theta_{\text{early}}\}. \quad (38)$$

In adverse-feasibility experiments we also report feasible top-risk coverage,

$$\text{FeasTopCov} = \frac{1}{|\mathcal{T}_\alpha|} \sum_{i \in \mathcal{T}_\alpha} \mathbf{1}\{C_i \leq \theta_{\text{early}}, Q_{s(i)} \leq B_d(1 - \rho)\}, \quad (39)$$

which prevents an early but energy-infeasible high-risk service from being counted as reliable coverage. For the selected service assignments $\mathcal{Y}$, the infeasible-sortie rate is

$$\text{IFR} = \frac{1}{|\mathcal{Y}|} \sum_{(s,i) \in \mathcal{Y}} \mathbf{1}\{Q_{si} > B_d(1 - \rho)\}. \quad (40)$$

Candidate-stop screening is evaluated by feasible-pair count and candidate-pair reduction, because these determine whether the service graph in Eq. (28) is rich enough for ALNS.

5.4 Statistical protocol and claim boundary

All main comparisons use matched random seeds where possible. Statistical testing uses paired Wilcoxon tests with Holm correction in the P7 block (Wilcoxon, 1945; Holm, 1979), pairing methods by seed, tower count, stop count and uncertainty level. The compact MILP/HiGHS reference uses the open-source HiGHS solver (Huangfu and Hall, 2018); it is used only for S-scale sanity checks and is not claimed as a complete branch-and-price benchmark. Public GIS cases are treated as GIS-grounded computational evidence, not utility field validation. AirLab telemetry validates that route/weather variables improve energy prediction on public quadcopter flights, not that the full inspection simulator has been field validated.

5.5 Experiment design map

Before reporting the results, Table 7 connects each result block to the specific model component, controlled factor and claim boundary it is meant to test. This table is included to avoid a common weakness in simulation papers: moving from setup directly to isolated figures without showing how each experiment defends a particular part of the formulation or algorithm.

Table 7: Experiment-design map connecting the model components, controlled comparisons and result evidence.

Result block	Question being tested	Model or algorithm link	Main control	Evidence artifact	Claim boundary
S-scale reference	Do the compact model and schedule evaluator operate on compatible scales?	Service graph; assignment; completion time	HiGHS vs ALNS variants	Table 8	Sanity/reference check only
Algorithm comparison	Does UQ/risk-value scheduling change priority performance beyond generic routing?	Objective Eq. (17); state Eq. (29)	Seven baseline families	Table 9; Fig. 4	Heuristic comparison under matched seeds
Energy evidence	Is the q95 energy gate supported by prediction and calibration evidence?	Eqs. (6)–(8)	Fixed/point/probabilistic and telemetry models	Table 10; Fig. 5	External UAV telemetry is calibration evidence, not field validation
Ablation/stress	Which components drive robust behavior under adverse feasibility?	q95 service graph; risk-value repair; ALNS operators	Component removals and stress cases	Tables 11–12; Fig. 6	Operator effects are bounded by the order-based implementation
Public GIS	Can the same pipeline be instantiated on public tower geometry/weather?	GIS-loaded $\mathcal{T}$, $\mathcal{S}$ and weather state	Region and baseline family	Table 13; Fig. 7	Public geometry plus proxy risk/value only
Scale and screening	How does Stage-1 graph density affect runtime, feasibility and risk-time?	Feasible set $\mathcal{F}$ in Eq. (28)	Tower count; stop mode; parameters	Tables 14–15; Fig. 9	Python-prototype runtime and controlled generator settings

6 Results and analysis

The results follow the evidence ladder summarized in Table 7. The first block checks whether the compact formulation and heuristic evaluator are numerically aligned on S-scale cases. The second block tests whether the risk-value objective in Eq. (17)

and the ALNS state in Eq. (29) improve inspection-priority metrics. The third block isolates the probabilistic energy interface in Eqs. (6)–(8). The remaining blocks then ask whether the same mechanism is robust under adverse feasibility, transferable to public GIS geometry and scalable when the candidate-stop graph is screened. Each result block is paired with a full raw-data table so that the reader can distinguish visual patterns from the underlying mean values, variances and evidence boundaries. This ordering is important: later figures are interpreted only after the model, energy and algorithm components they depend on have been tested.

6.1 Small reference and evidence boundary

Before interpreting the large-instance heuristic results, the compact reference checks whether the mathematical model and the schedule evaluator describe the same operational scale. Table 8 reports the raw S-scale means used for this check. On 10-, 15- and 20-tower cases, the compact HiGHS reference and the ALNS variants produce schedules on similar makespan scales: the mean makespans for the proposed method are 96.81, 66.23 and 90.22, compared with 98.35, 66.18 and 89.17 for the compact MILP reference. The proposed method also removes residual infeasible sorties in the 15- and 20-tower cases and keeps the 10-tower infeasible-sortie rate at 0.01. This block is not an optimality claim, because the compact reference does not optimize RWCT and does not include every synchronization and risk-value feature used by the heuristic evaluator. Its role is narrower: it checks that coverage, service-pair feasibility and completion-time accounting remain plausible before the paper moves to M- and L-scale corridor experiments.

Table 8: S-scale compact-reference diagnostics. The HiGHS row optimizes the compact makespan–energy model; RWCT and top-risk coverage are computed afterwards by the common post-hoc schedule evaluator and are not HiGHS objective values. Values are means over matched seeds; shaded rows denote the proposed configuration.

Towers	Stops	Method	Makespan	Eval. RWCT	Eval. top cov.	Infeas.	Time (s)
10	5	Compact HiGHS (Cmax ref.)	98.35	15414	0.350	0.070	0.013
10	5	Point ALNS-PINN	96.01	9673	0.850	0.070	0.001
10	5	Proposed	96.81	9699	0.800	0.010	0.004
15	8	Compact HiGHS (Cmax ref.)	66.18	17722	0.300	0.000	0.082
15	8	Point ALNS-PINN	67.03	11001	0.867	0.000	0.002
15	8	Proposed	66.23	10907	0.833	0.000	0.009
20	10	Compact HiGHS (Cmax ref.)	89.17	31868	0.340	0.000	0.104
20	10	Point ALNS-PINN	89.89	21161	0.900	0.000	0.003
20	10	Proposed	90.22	20986	0.920	0.000	0.012

6.2 Algorithm comparison

The main routing question is whether the inspection-specific terms added in the model change scheduling behavior beyond generic order search. Table 9 gives the full M/L-scale comparison across all tested tower counts and baselines, while Fig. 4 focuses on the 100- and 500-tower cases to make the main pattern visible. On the 100-tower benchmark, the proposed method reduces RWCT by 34.4% relative to fixed ALNS and by 0.76% relative to point ALNS-PINN. The paired RWCT comparison against point ALNS-PINN has a median paired reduction of 639.3 risk-time units, a bootstrap median-difference interval of 499.1–759.6 and a Holm-adjusted Wilcoxon value of $p_{Holm} = 0.0117$. Top-risk coverage improves descriptively by 2.4 percentage points; its unadjusted paired value is $p = 0.039$, but after the same Holm family correction it is $p_{Holm} = 0.157$. These comparisons answer different parts of the method design. The large gap over fixed ALNS supports the move away from fixed endurance and cost-only priority. The smaller gap over point ALNS-PINN shows the narrower but still measurable value of upper-quantile screening and the full risk-value configuration once mean-energy prediction has already improved ordering. The proposed method accepts a slightly larger nominal makespan because it prioritizes conservative feasibility and earlier service of high-risk towers.

Table 9: Full P2 medium-uncertainty algorithm-comparison data across all tested corridor scales; values are means over ten matched seeds. Shaded rows denote the proposed configuration; bold values indicate the best value within each tower-count block for the corresponding metric direction.

Scale	Towers	Method	Makespan ↓	RWCT ↓	Top cov. ↑	Infeas. ↓	Time (s) ↓
M	30	Nearest	63.99	28419	0.243	0.0000	0.001
M	30	GA	63.32	18679	0.857	0.0000	0.019
M	30	ACO	63.01	18676	0.857	0.0000	0.036
M	30	Fixed ALNS	64.74	27694	0.214	0.0000	0.019
M	30	Point ALNS-PINN	62.94	18715	0.857	0.0000	0.006
M	30	UQ ALNS-PINN	62.81	18718	0.857	0.0000	0.006
M	30	Proposed	63.61	18581	0.871	0.0000	0.023
M	50	Nearest	105.38	74345	0.292	0.0080	0.002
M	50	GA	104.96	52279	0.833	0.0080	0.043
M	50	ACO	105.03	52256	0.825	0.0080	0.078

Table 9: Full P2 medium-uncertainty algorithm-comparison data across all tested corridor scales (continued).

Scale	Towers	Method	Makespan ↓	RWCT ↓	Top cov. ↑	Infeas. ↓	Time (s) ↓
M	50	Fixed ALNS	106.31	73506	0.258	0.0080	0.062
M	50	Point ALNS-PINN	105.11	52332	0.825	0.0080	0.020
M	50	UQ ALNS-PINN	104.97	52383	0.833	0.0000	0.019
M	50	Proposed	106.05	51911	0.892	0.0000	0.077
M	75	Nearest	72.60	73283	0.250	0.0013	0.004
M	75	GA	71.62	47588	0.872	0.0013	0.086
M	75	ACO	71.52	47572	0.867	0.0013	0.139
M	75	Fixed ALNS	72.74	71201	0.239	0.0013	0.133
M	75	Point ALNS-PINN	71.44	47628	0.872	0.0013	0.042
M	75	UQ ALNS-PINN	71.76	47639	0.872	0.0000	0.044
M	75	Proposed	71.68	47287	0.889	0.0000	0.177
M	100	Nearest	93.54	130448	0.260	0.0020	0.006
M	100	GA	92.89	84649	0.896	0.0020	0.133
M	100	ACO	92.48	84608	0.896	0.0020	0.200
M	100	Fixed ALNS	93.74	128168	0.240	0.0020	0.232
M	100	Point ALNS-PINN	92.69	84732	0.896	0.0020	0.072
M	100	UQ ALNS-PINN	92.91	84787	0.896	0.0000	0.072
M	100	Proposed	93.39	84091	0.920	0.0000	0.294
L	150	Nearest	202.65	401903	0.254	0.0027	0.011
L	150	GA	200.74	256630	0.870	0.0027	0.214
L	150	ACO	201.11	256393	0.868	0.0027	0.302
L	150	Fixed ALNS	202.56	395003	0.257	0.0027	0.470
L	150	Point ALNS-PINN	200.77	257090	0.876	0.0027	0.154
L	150	UQ ALNS-PINN	200.86	257425	0.876	0.0000	0.158
L	150	Proposed	200.77	254062	0.908	0.0000	0.632
L	200	Nearest	138.93	367151	0.216	0.0040	0.021
L	200	GA	136.47	229511	0.886	0.0040	0.375
L	200	ACO	136.70	229299	0.880	0.0040	0.481
L	200	Fixed ALNS	138.58	357117	0.226	0.0040	0.862
L	200	Point ALNS-PINN	136.46	229884	0.888	0.0040	0.273
L	200	UQ ALNS-PINN	136.80	230123	0.890	0.0000	0.281
L	200	Proposed	137.95	227561	0.912	0.0000	1.131
L	300	Nearest	165.80	654195	0.257	0.0010	0.044
L	300	GA	163.48	412351	0.889	0.0010	0.758
L	300	ACO	163.43	412045	0.884	0.0010	0.915
L	300	Fixed ALNS	165.38	641222	0.260	0.0010	1.894
L	300	Point ALNS-PINN	163.24	413088	0.888	0.0010	0.633
L	300	UQ ALNS-PINN	163.18	413332	0.888	0.0000	0.618
L	300	Proposed	164.93	408886	0.924	0.0000	2.480
L	500	Nearest	204.90	1283058	0.249	0.0032	0.112
L	500	GA	203.94	810172	0.883	0.0032	1.918
L	500	ACO	203.74	809488	0.886	0.0032	2.155
L	500	Fixed ALNS	204.53	1256405	0.245	0.0032	5.249
L	500	Point ALNS-PINN	203.74	811648	0.884	0.0032	1.698
L	500	UQ ALNS-PINN	203.65	812385	0.884	0.0000	1.704
L	500	Proposed	206.00	803530	0.919	0.0000	6.616

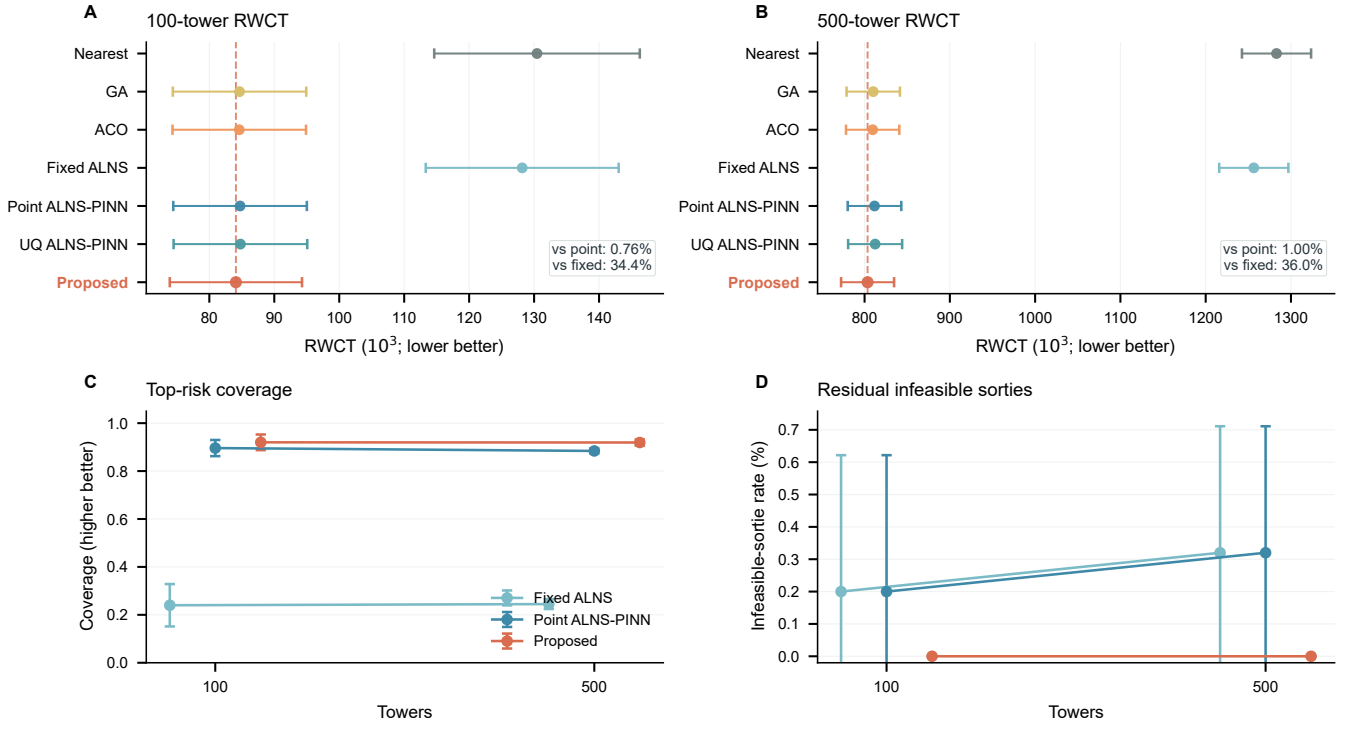


Figure 4: Algorithm-comparison evidence under matched 100- and 500-tower seeds. A–B: mean risk-weighted completion time with standard-deviation bars for all baselines; dashed vertical lines mark the proposed mean. C: top-risk coverage for the main ALNS variants. D: residual infeasible-sortie rate. Lower is better in A, B and D, whereas higher is better in C; Table 9 reports the raw values.

The L-scale results follow the same logic. At 500 towers, the proposed method reduces RWCT by 36.1% relative to fixed ALNS and by 1.0% relative to point ALNS-PINN. Against the point variant, the median paired RWCT reduction is 8026 risk-time units with a bootstrap median-difference interval of 7695–9337 and $p_{Holm} = 0.0117$. Top-risk coverage increases from 0.884 to 0.919, with a median paired gain of 0.032 and $p_{Holm} = 0.0117$, while the infeasible-sortie rate falls from 0.0032 to zero. The main interpretation is therefore not that the proposed method always minimizes makespan. It changes the tradeoff toward lower risk-weighted delay and fewer residual energy violations, which is exactly the role assigned to $\psi_i C_i$ and Q_{si} in the formulation.

6.3 Energy-prediction evidence

The routing gains rely on the energy interface being more than a cosmetic predictor. Table 10 and Fig. 5 therefore separate scheduler-facing simulation evidence from external telemetry calibration. In the simulation prediction task, the probabilistic PINN surrogate reduces high-uncertainty MAE from 12.96 for fixed physics and 9.40 for point PINN to 1.30, while increasing 95% coverage to 1.00 and eliminating false-feasible decisions in the high-uncertainty block. In the low- and medium-uncertainty blocks, the residual false-feasible rate is 0.0020, so the supported general claim is reduction rather than universal elimination. This supports the use of Q_{si} in the service graph. In the public AirLab telemetry calibration, the best telemetry-weather linear model reduces held-out MAE from 3.902 Wh for the constant baseline to 0.890 Wh. Because percentage error can be unstable for low-energy flights, Table 10 also reports WAPE and SMAPE: the telemetry-weather model gives the lowest WAPE (0.043), whereas the route-linear model has the lowest SMAPE (0.071) but higher MAE and RMSE. That second block supports the premise that route and weather variables matter in real quadcopter energy data; it does not validate the full inspection simulator in the field.

Table 10: Full energy-prediction data for scheduler-facing simulation and AirLab telemetry calibration. AirLab rows report MAPE, WAPE and SMAPE because MAPE is unstable for low-energy flights. Shaded rows denote the selected probabilistic or telemetry-weather model; bold values indicate the best value within each uncertainty level or telemetry block.

Block	Level	Model	Test	MAE ↓	RMSE ↓	MAPE ↓	WAPE ↓	SMAPE ↓	Cov. ↑	False feas. ↓
Simulation	Low	Fixed physics	10 seeds	12.03	12.86	0.188	—	—	0.132	0.0040
Simulation	Low	Point PINN	10 seeds	8.51	9.08	0.135	—	—	0.516	0.0020
Simulation	Low	Prob. PINN	10 seeds	0.91	1.15	0.015	—	—	1.000	0.0020
Simulation	Medium	Fixed physics	10 seeds	12.37	13.23	0.192	—	—	0.118	0.0040
Simulation	Medium	Point PINN	10 seeds	8.84	9.45	0.140	—	—	0.484	0.0020
Simulation	Medium	Prob. PINN	10 seeds	1.05	1.32	0.017	—	—	0.998	0.0020
Simulation	High	Fixed physics	10 seeds	12.96	13.87	0.200	—	—	0.102	0.0040
Simulation	High	Point PINN	10 seeds	9.40	10.12	0.147	—	—	0.430	0.0020
Simulation	High	Prob. PINN	10 seeds	1.30	1.64	0.021	—	—	1.000	0.0000
AirLab	Held-out	Constant mean	41 flights	3.902 Wh	5.316 Wh	3.281	0.188	0.209	0.976	0.0000
AirLab	Held-out	Parameter linear	41 flights	2.410 Wh	3.250 Wh	2.112	0.116	0.147	0.976	0.0000
AirLab	Held-out	Route linear	41 flights	1.260 Wh	1.544 Wh	0.075	0.061	0.071	0.976	0.0000
AirLab	Held-out	Telemetry-weather	41 flights	0.890 Wh	1.083 Wh	0.067	0.043	0.091	1.000	0.0000

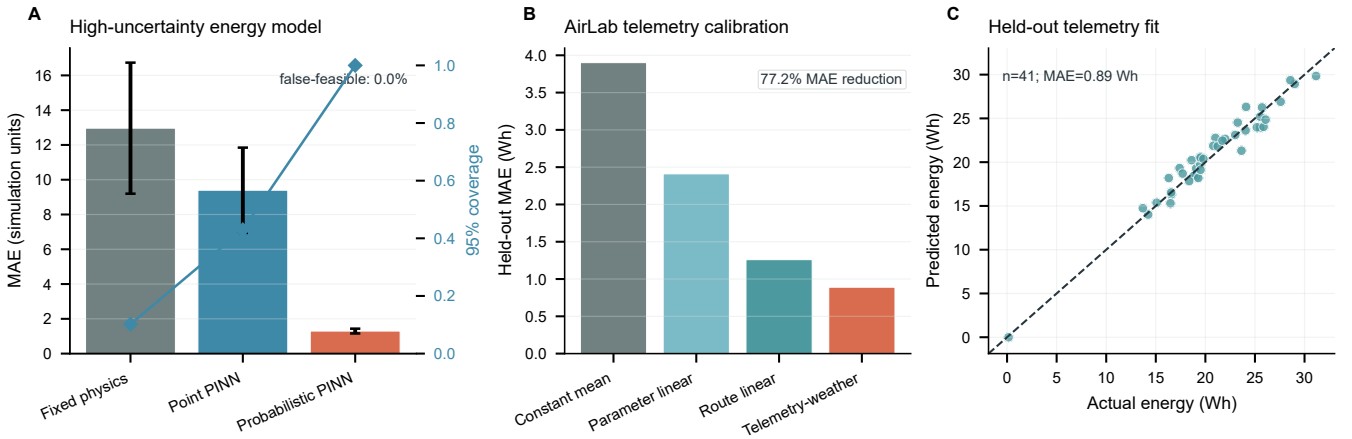


Figure 5: Energy-prediction evidence supporting the q95 service graph. A: high-uncertainty simulation MAE with standard-deviation bars and 95% coverage on the right axis; B: held-out AirLab telemetry MAE for progressively richer calibration models; C: actual versus predicted held-out energy for the telemetry-weather model. The telemetry block is external calibration evidence, not field validation of transmission-line inspection sorties.

6.4 Ablation, stress tests and operator interpretation

The ablation block first removes individual modeling and search components on the 50-tower high-uncertainty case (Table 11). Removing the energy predictor or the risk-value term changes the inspection-priority behavior most strongly. Removing UQ gives a slightly lower nominal RWCT in this batch, but it reintroduces residual infeasible sorties and lowers feasible top-risk coverage from 0.892 to 0.883. Removing only the named energy-minimum or synchronization-aware repair operator has limited final effect in this implementation. The stress tests then connect those components to the adverse settings that motivated them: sparse feasible stops, high wind and tight battery reserve. Under sparse high-wind, tight-battery and very sparse-corridor cases, the full framework reduces RWCT relative to fixed ALNS by about 23.6%, 26.0% and 25.4%, respectively (Table 12). Removing UQ changes the tradeoff sharply: nominal top-risk coverage rises to 0.611–0.756 in the stress cases, but q95-feasible coverage falls to 0.233–0.312 and the infeasible-sortie rate stays near 0.73–0.75. The supported claim is therefore deliberately bounded: the broad UQ/risk-value framework improves the reliability side of adverse scheduling, while the two named repair operators are useful search components and diagnostics rather than independently proven final-quality drivers.

Table 11: Full ablation data on the 50-tower high-uncertainty case. Shaded rows denote the proposed configuration; bold values indicate the best value within the ablation block.

Method	Stop mode	Makespan ↓	RWCT ↓	Top/feas. cov. ↑	Infeas. ↓	Time (s) ↓	Acc.	Imp.
No PINN	kmeans	106.19	73455	0.258/0.258	0.020	0.051	88.8	20.3
No UQ	kmeans	104.95	51886	0.892/0.883	0.018	0.067	76.9	17.0
No risk-value	kmeans	106.99	73561	0.258/0.258	0.000	0.066	89.3	18.8
No adaptive	kmeans	106.21	51917	0.892/0.892	0.000	0.067	59.7	25.4
No clustering	direct	117.69	57987	0.900/0.900	0.000	0.065	76.5	18.2
No energy repair	kmeans	106.08	51908	0.892/0.892	0.000	0.070	87.1	16.2
No sync repair	kmeans	106.12	51908	0.892/0.892	0.000	0.068	85.2	14.9
Proposed	kmeans	106.16	51910	0.892/0.892	0.000	0.067	77.2	20.8

Table 12: Full stress-test data for uncertainty and repair components. SHW: sparse high wind; TB: tight battery; VSC: very sparse corridor. Top/feas. cov. reports nominal early top-risk coverage followed by q95-feasible early top-risk coverage. Shaded rows denote the proposed configuration; bold values indicate the best value within each stress case.

Case	Method	Makespan ↓	RWCT ↓	Top/feas. cov. ↑	Infeas. ↓	Time (s) ↓	Acc.	Imp.
SHW	Fixed ALNS	355.34	267139	0.222/0.217	0.745	0.070	125.6	23.0
SHW	No UQ	352.86	202441	0.611/0.233	0.745	0.086	111.4	23.8
SHW	No risk-value	357.99	270191	0.222/0.222	0.555	0.086	126.2	19.6
SHW	No energy repair	357.86	204099	0.583/0.383	0.555	0.088	123.6	19.6
SHW	No sync repair	357.86	204099	0.583/0.383	0.555	0.086	123.9	15.6
SHW	Proposed	357.84	204105	0.583/0.383	0.555	0.086	110.9	21.0
TB	Fixed ALNS	241.14	179414	0.244/0.233	0.741	0.082	126.1	20.1
TB	No UQ	240.46	130609	0.756/0.267	0.741	0.102	111.9	20.1
TB	No risk-value	244.30	183300	0.239/0.239	0.465	0.103	126.5	19.8
TB	No energy repair	243.16	132691	0.717/0.478	0.465	0.100	124.3	20.4
TB	No sync repair	243.16	132691	0.717/0.478	0.465	0.102	122.4	14.8
TB	Proposed	243.16	132691	0.717/0.478	0.465	0.109	112.5	20.6
VSC	Fixed ALNS	749.51	663245	0.288/0.288	0.732	0.085	126.3	24.2
VSC	No UQ	744.55	492583	0.604/0.312	0.731	0.103	112.9	24.3
VSC	No risk-value	749.96	666369	0.288/0.288	0.646	0.102	126.3	23.1
VSC	No energy repair	746.53	494617	0.556/0.376	0.646	0.104	124.2	19.8
VSC	No sync repair	746.53	494617	0.556/0.376	0.646	0.104	123.7	16.0
VSC	Proposed	746.53	494617	0.556/0.376	0.646	0.103	109.9	20.0

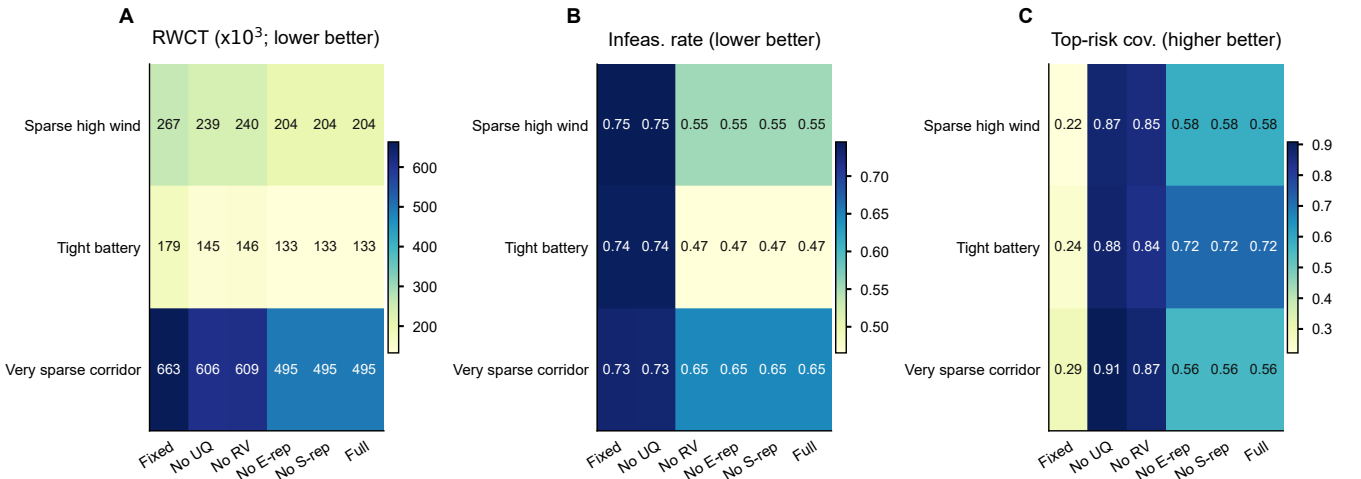


Figure 6: Stress-test heatmaps for adverse feasibility cases. A: risk-weighted completion time; B: infeasible-sortie rate; C: nominal top-risk coverage. Rows and columns are shared across panels, but each panel uses its own colorbar because the metrics have different units and favorable directions. Lower is better in A and B, whereas higher is better in C; Table 12 pairs the nominal coverage values with q95-feasible top-risk coverage.

6.5 Public GIS-grounded cases

After the synthetic and stress evidence, the public GIS block asks whether the same service-graph construction can be instantiated on real public tower geometry and weather inputs. Figure 7 maps the Bay Area, Los Angeles inland and Dallas–Fort Worth cases

in projected local coordinates, with towers colored by proxy risk-value and candidate stops shown as support positions. Table 13 reports all baseline rows for the three regions. The full method has the lowest RWCT in all three cases, but the margins are small in Bay Area and Los Angeles, where the best non-full baseline is ACO and the reduction is below 0.5%. Dallas–Fort Worth has the strongest RWCT signal, with a 3.262% reduction over the best non-full baseline, but the metric-alignment audit shows that it is not a pure high-risk-coverage win: point and UQ ALNS-PINN reach TopCov 0.933, while the proposed row is 0.867, and the nearest baseline has the second-best RWCT despite TopCov 0. This case is therefore used as evidence that the proxy risk-value objective changes aggregate delay, not as a stand-alone proof of superior early high-risk coverage. Because risk, value and stops are generated proxies, these cases are GIS-grounded computational evidence rather than utility field validation.

Table 13: Full public GIS-grounded case data. Geometry and weather are public data; risk/value labels remain proxies. Shaded rows denote the proposed configuration; bold values indicate the best value within each region.

Case	Method	Makespan ↓	RWCT ↓	Top cov. ↑	Infeas. ↓	Time (s) ↓
Bay Area	Nearest	85.01	148681	0.400	0.000	0.002
Bay Area	GA	80.85	139311	0.933	0.000	0.042
Bay Area	ACO	80.85	139296	0.933	0.000	0.071
Bay Area	Fixed ALNS	84.52	153610	0.200	0.000	0.061
Bay Area	Point ALNS-PINN	81.36	139503	0.933	0.000	0.020
Bay Area	UQ ALNS-PINN	81.64	139515	0.933	0.000	0.020
Bay Area	Proposed	81.62	139230	1.000	0.000	0.083
Dallas–Fort Worth	Nearest	117.31	178676	0.000	0.117	0.002
Dallas–Fort Worth	GA	107.22	189612	0.867	0.117	0.043
Dallas–Fort Worth	ACO	106.57	188897	0.867	0.117	0.074
Dallas–Fort Worth	Fixed ALNS	116.47	181359	0.000	0.117	0.065
Dallas–Fort Worth	Point ALNS-PINN	106.94	190267	0.933	0.117	0.021
Dallas–Fort Worth	UQ ALNS-PINN	108.09	190730	0.933	0.100	0.020
Dallas–Fort Worth	Proposed	113.65	172848	0.867	0.100	0.087
Los Angeles inland	Nearest	91.89	164184	0.133	0.000	0.002
Los Angeles inland	GA	93.07	158087	0.933	0.000	0.043
Los Angeles inland	ACO	93.12	158074	0.933	0.000	0.075
Los Angeles inland	Fixed ALNS	92.09	167532	0.133	0.000	0.063
Los Angeles inland	Point ALNS-PINN	92.91	158458	0.933	0.000	0.022
Los Angeles inland	UQ ALNS-PINN	92.74	158542	0.933	0.000	0.020
Los Angeles inland	Proposed	93.10	157353	0.933	0.000	0.082

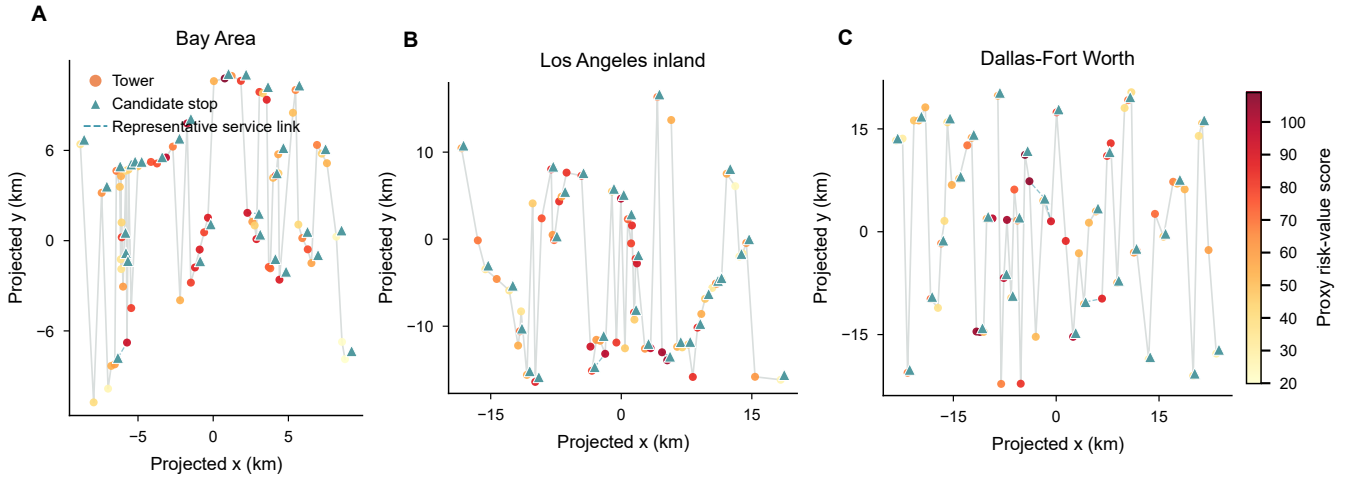


Figure 7: Public GIS-grounded cases in projected local coordinates. Towers are colored by proxy risk-value score; triangles denote generated candidate stops; dashed links show representative high-risk service relations used to visualize the service graph. The coordinates and weather inputs are public-data grounded, whereas risk/value labels and stops remain generated proxies.

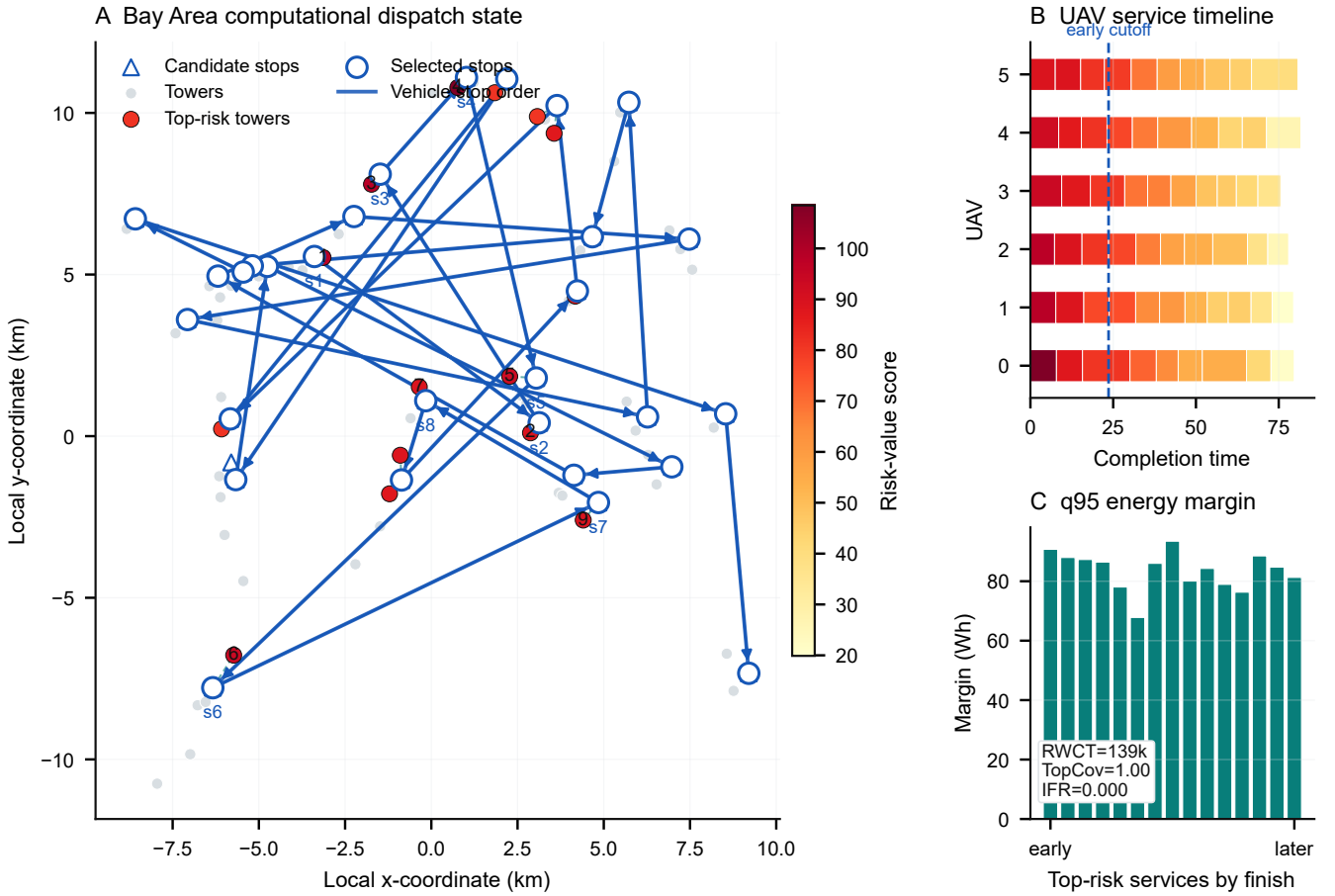


Figure 8: Representative operating-state diagnostic for the Bay Area public-GIS case solved by the proposed configuration. A: selected stops, vehicle stop order and high-risk UAV service relations in local projected coordinates; B: UAV service timeline colored by tower risk-value score, with the early-service cutoff used in top-risk coverage; C: q95 energy margin for early high-risk services. This figure is a computational dispatch trace from the solver, not a field-operation log.

Figure 8 adds the operating-state view that the aggregate GIS table cannot show. The Bay Area run serves the highest-risk quartile before the early cutoff and retains positive q95 energy margins for the displayed high-risk services, which is consistent with the zero infeasible-sortie rate reported by the proposed row in Table 13. The diagnostic is deliberately tied to the same PlanTask records used by the metric evaluator; it should therefore be read as an internal schedule trace for checking route, timeline and energy-margin consistency.

6.6 Scalability and candidate-stop screening

The final result block returns to Stage 1 of the algorithm: the candidate-stop generator controls the density of the service graph in Eq. (28). Table 14 gives the raw scale, screening and UAV-count values behind Fig. 9; Table 15 reports the additional parameter-sensitivity rows for candidate mode, iteration budget, quantile coefficient, reserve ratio and UAV count. K-means screening removes 80.0% of candidate pairs at 30 towers and 94.0% at 100 towers while preserving zero infeasible sorties in the tested corridor generator. Direct enumeration still reports the full screened graph, but its q95-feasible subset is much smaller than the raw stop-tower product: 55.9 feasible pairs out of 450 total pairs at 30 towers and 204.1 out of 5000 at 100 towers. DBSCAN leaves only about 12–13 feasible pairs across these M-scale cases, which is too sparse for reliable scheduling and leads to infeasible-sortie rates from 0.37 to 0.81. Increasing the UAV count from two to six reduces makespan sharply, indicating that fleet size is an operationally important design parameter. The runtime curve remains practical in the current Python prototype, but these absolute runtimes should not be interpreted as optimized production solver times.

Table 14: Raw scalability, candidate-stop screening and UAV-count sensitivity data used in Fig. 9; values are means over ten matched seeds. Scale/screening rows use the medium-uncertainty P2/P6 setting, whereas UAV-count rows come from the high-uncertainty P8 sensitivity block. Feasible pairs are q95-feasible stop-tower pairs, including direct enumeration. Top/feas. cov. reports nominal and q95-feasible early top-risk coverage. Shaded rows denote the K-means screened proposed configuration or the best UAV-count setting in the sensitivity block.

Block	Setting	Towers	Stops	Pair red.	Feas. pairs	RWCT	Top/feas. cov.	Infeas.	Time (s)
Scale	K-means	30	15	80.0%	55.5	18581	0.871/0.871	0.000	0.023
Scale	K-means	50	25	88.1%	93.9	51911	0.892/0.892	0.000	0.077
Scale	K-means	75	35	91.4%	135.7	47287	0.889/0.889	0.000	0.177
Scale	K-means	100	50	94.0%	200.3	84091	0.920/0.920	0.000	0.294
Scale	K-means	150	75	96.0%	290.7	254062	0.908/0.908	0.000	0.632
Scale	K-means	200	100	97.0%	398.7	227561	0.912/0.912	0.000	1.131
Scale	K-means	300	150	98.0%	625.1	408886	0.924/0.924	0.000	2.480
Scale	K-means	500	250	98.8%	964.5	803530	0.919/0.919	0.000	6.616
Screening	Direct	30	15	0.0%	55.9	21684	0.886/0.886	0.000	0.028
Screening	DBSCAN	30	15	38.2%	12.8	41318	0.686/0.457	0.367	0.012
Screening	K-means	30	15	80.0%	55.5	18581	0.871/0.871	0.000	0.029
Screening	Direct	50	25	0.0%	96.5	57953	0.900/0.900	0.000	0.066
Screening	DBSCAN	50	25	50.7%	12.0	150481	0.700/0.375	0.622	0.017
Screening	K-means	50	25	88.1%	93.9	51911	0.892/0.892	0.000	0.066
Screening	Direct	75	35	0.0%	136.3	54823	0.900/0.900	0.024	0.130
Screening	DBSCAN	75	35	55.6%	12.5	187700	0.689/0.272	0.763	0.029
Screening	K-means	75	35	91.4%	135.7	47286	0.889/0.889	0.000	0.134
Screening	Direct	100	50	0.0%	204.1	94073	0.920/0.920	0.002	0.234
Screening	DBSCAN	100	50	58.1%	12.8	431643	0.676/0.188	0.809	0.039
Screening	K-means	100	50	94.0%	200.3	84091	0.920/0.920	0.000	0.231
UAV count	2 UAVs	50	25	88.1%	78.8	97246	0.917/0.917	0.000	0.071
UAV count	4 UAVs	50	25	88.1%	78.8	51908	0.892/0.892	0.000	0.071
UAV count	6 UAVs	50	25	88.1%	78.8	36813	0.892/0.892	0.000	0.073

Table 15: Full sensitivity-analysis data on the 50-tower high-uncertainty case. Shaded rows denote the default proposed setting used in the main experiments; bold values indicate the best value within each sensitivity factor.

Factor	Level	Makespan ↓	RWCT ↓	Top cov. ↑	Infeas. ↓	Time (s) ↓
Stop mode	direct	117.76	57978	0.900	0.000	0.072
Stop mode	kmeans	106.12	51908	0.892	0.000	0.071
Stop mode	dbscan	326.04	149489	0.700	0.714	0.018
Iterations	50	106.40	51920	0.892	0.000	0.069
Iterations	100	106.12	51908	0.892	0.000	0.073
Iterations	160	106.12	51908	0.892	0.000	0.077
$z_{\mathcal{E}}$	1.28	105.87	51937	0.892	0.000	0.070
$z_{\mathcal{E}}$	1.645	106.12	51908	0.892	0.000	0.069
$z_{\mathcal{E}}$	1.96	105.41	51908	0.875	0.004	0.072
ρ	0.08	106.10	51947	0.892	0.000	0.076
ρ	0.12	106.12	51908	0.892	0.000	0.073
ρ	0.2	105.56	51910	0.883	0.006	0.072
UAV count	2	203.20	97246	0.917	0.000	0.071
UAV count	4	106.12	51908	0.892	0.000	0.071
UAV count	6	73.84	36813	0.892	0.000	0.073

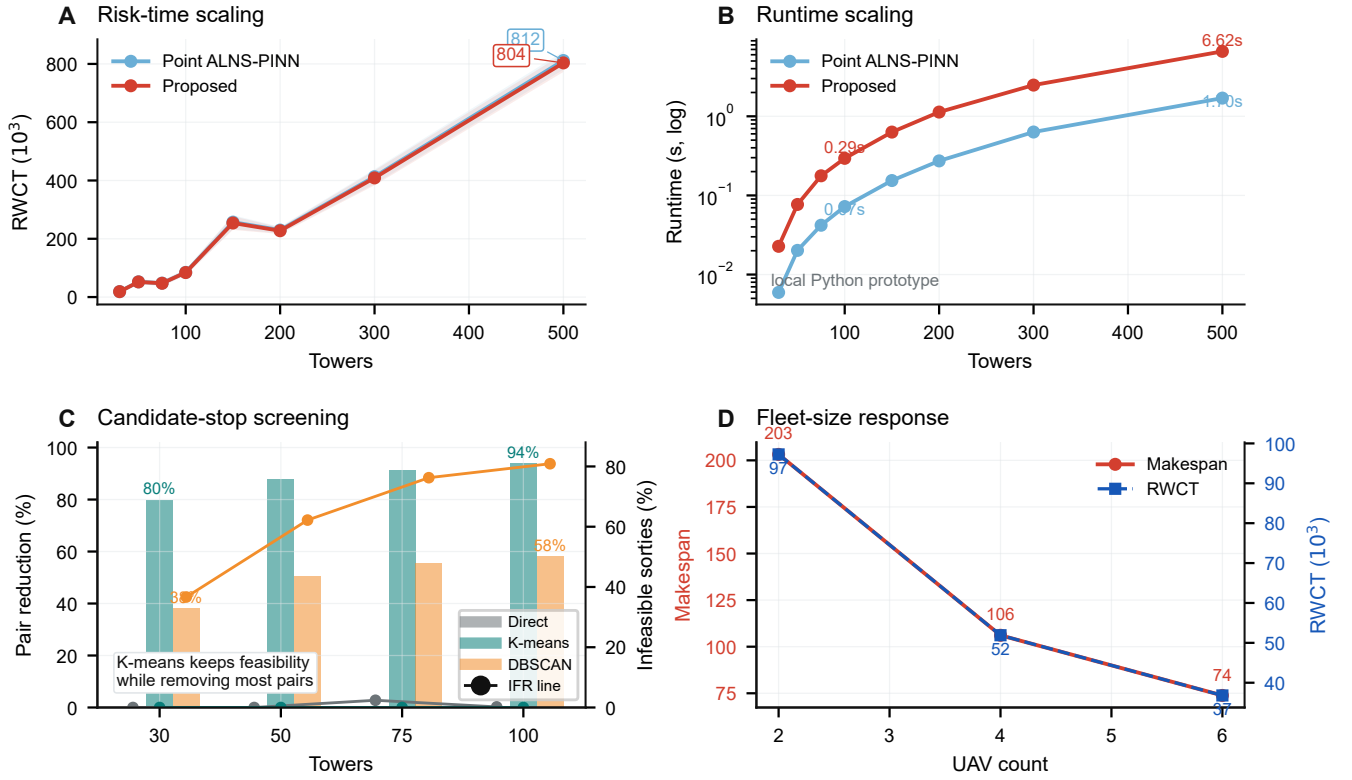


Figure 9: Scalability and screening trade-offs for the Stage-1 service graph. A–B: risk-time and runtime scaling of point-estimate ALNS-PINN and the proposed method with value labels at representative endpoints; C: candidate-pair reduction and infeasible-sortie rate for direct, K-means and DBSCAN stop screening; D: UAV-count sensitivity with makespan and RWCT shown on separate axes. Table 14 reports the raw data behind all panels.

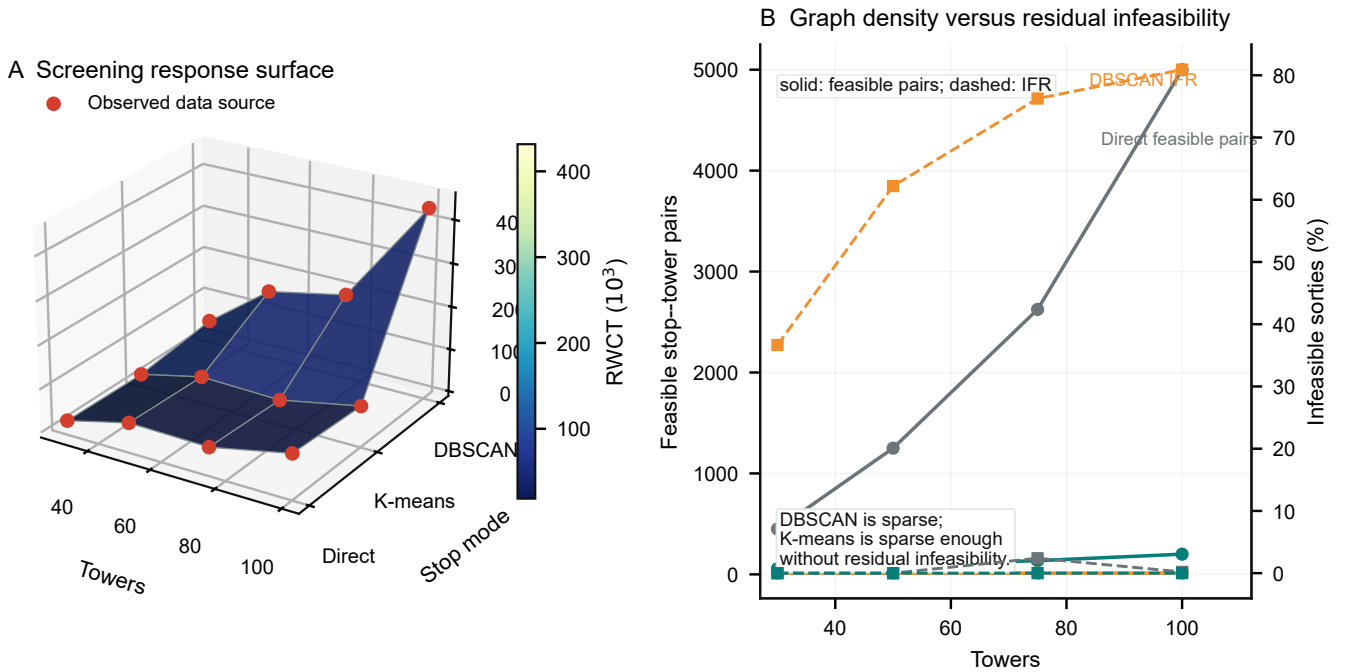


Figure 10: Candidate-stop screening response surface built from the observed screening experiment in Table 14. A: RWCT surface over tower count and categorical stop-generation mode, with red markers denoting the original data points; B: feasible stop-tower pairs and residual infeasible-sortie rate across the same conditions. The mode axis is categorical, so the surface should be read as a compact visualization of the observed grid rather than a continuous physical interpolation.

Figure 10 makes the graph-density mechanism explicit. Direct enumeration preserves all candidate pairs for evaluation, but the feasible-pair column counts only q95-admissible stop-tower edges. DBSCAN creates a very sparse feasible graph and the infeasible-sortie rate rises sharply. K-means lies between those extremes: it removes most candidate pairs while preserving enough feasible edges for the tested generator. This explains why K-means is used as the default candidate-stop screening mode in Table 4

rather than being treated as an arbitrary preprocessing choice.

7 Discussion

The evidence supports three main conclusions. First, risk-value scheduling changes the optimization target in a way that is relevant to inspection. The proposed method is not always the fastest in nominal makespan, but Tables 9 and 12 show lower risk-weighted completion time or a more reliable feasible-coverage tradeoff than fixed ALNS across the main large-scale and stress tests. Second, probabilistic energy screening converts energy prediction into a routing decision. The $q_{0.95}$ filter removes residual infeasible sorties in the 100- and 500-tower comparisons, while Table 10 shows that route and weather variables can materially reduce held-out energy error in telemetry calibration. Third, candidate-stop generation is a substantial computational layer, not a preprocessing detail. Figure 9 shows that screening changes the feasible service-pair set, and overly aggressive screening can dominate the final schedule.

The main limitations are also clear. The MILP reference is compact and used only for small instances. The public GIS cases use real tower geometries and weather but not real utility defect labels or field crew schedules. The AirLab telemetry data are real quadcopter flights, but they are package-delivery flights rather than transmission-line inspection sorties. The ALNS repair operators are implemented and tested, but the current order-only state representation limits their independent effect on final aggregate metrics.

8 Conclusions

This paper develops an uncertainty-aware risk-value vehicle–UAV scheduling framework for transmission-line inspection. The method combines probabilistic energy prediction, chance-style sortie screening, candidate-stop generation and ALNS-based routing. The computational evidence follows a TRE-style structure: formulation, algorithm, benchmark comparison, energy validation, ablation, sensitivity and GIS-grounded cases. The proposed method improves risk-weighted completion time and top-risk coverage relative to fixed ALNS and point-estimate ALNS-PINN in the main large-scale comparisons, while reducing residual infeasible-sortie rates. Stress cases show a reliability tradeoff rather than universal dominance on nominal coverage. Public GIS-grounded cases and AirLab telemetry provide useful external grounding, but the manuscript should not claim utility field validation. Future work should incorporate real inspection logs, defect labels, field battery records and a fuller exact optimization benchmark for synchronization-rich small instances.

Data availability

The computational results are generated from reproducible simulation scripts and CSV source data. Public-source inputs include OpenStreetMap tower nodes, HIFLD transmission-line context, NASA POWER weather and the AirLab quadcopter telemetry dataset. The synthetic instance data, experiment outputs and figure source data can be made available with the implementation package.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

No acknowledgements are included in the review version.

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